

Advanced Cognitive Neuroscience

Week 40: Forward modelling, inverse modelling (and dipole estimation)

The course plan

Week 37:

Monday: Workshop paradigm: Measuring visual subjective experience
MR Recordings (Lau)

Week 38:

Monday: What is it all about? (Lau)
Workshop 1: MR-recordings (Laura & Lau)

Week 39:

Monday: Basic physiology and Evoked responses (Lau)
Wednesday: Preprocessing and evoked responses (Laura)

Week 40:

Monday: Forward modelling and inverse modelling (Lau)
Wednesday: Source reconstruction (Laura)

Week 41:

Monday: Multivariate statistics (Laura) (Lecture)
Wednesday: Multivariate statistics (Laura) (Coding)

Deadline for feedback: Video Explainer

Week 43:

Monday: Inverse modelling: beamforming (Lau) (Lecture)
Wednesday: Inverse modelling: beamforming (Laura) (Coding)

Week 44:

Monday From single subject analysis to group analysis (Laura)
Wednesday: Work on your reports (Laura)

Week 45:

Monday: What about that other cortex? - the cerebellar one (Lau)
Wednesday: Work on your reports, continued (Laura)

Week 46:

Monday: Optically pumped magnetoencephalography (Laura)
Wednesday: Respiration (Laura)

Deadline for feedback: Lab report

Week 47:

No teaching: prepare presentations!

Week 48:

Monday: Oral presentations (part 1) (Lau)
Wednesday: Oral presentations (part 2) (Lau)

Week 49:

Monday: Oral presentations (part 3) (Lau)
Wednesday: Oral presentations (part 4) (Lau)

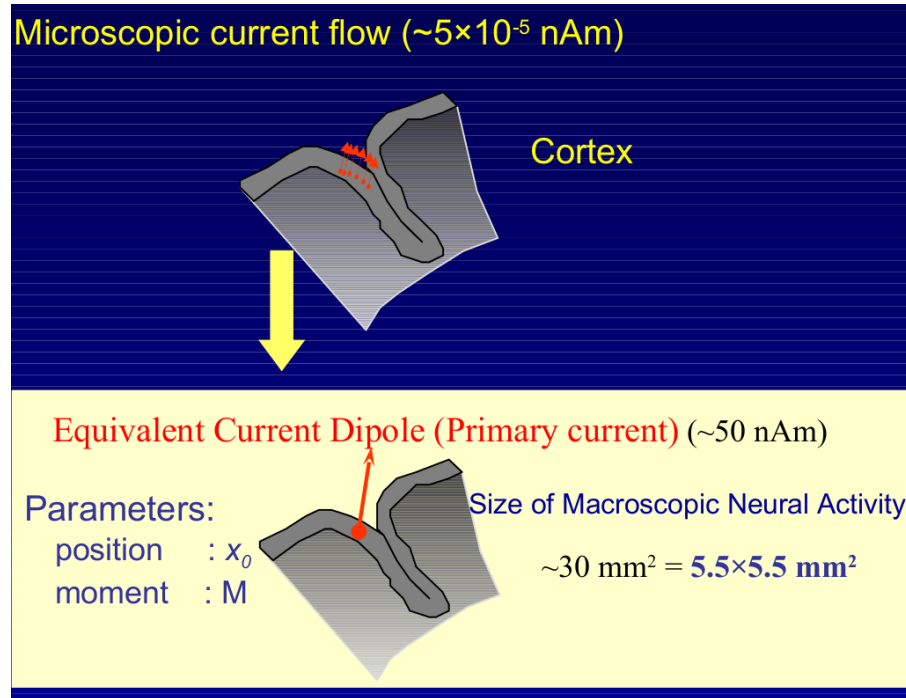
Week 50:

Wednesday: What was it all about (Lau)

Learning goals

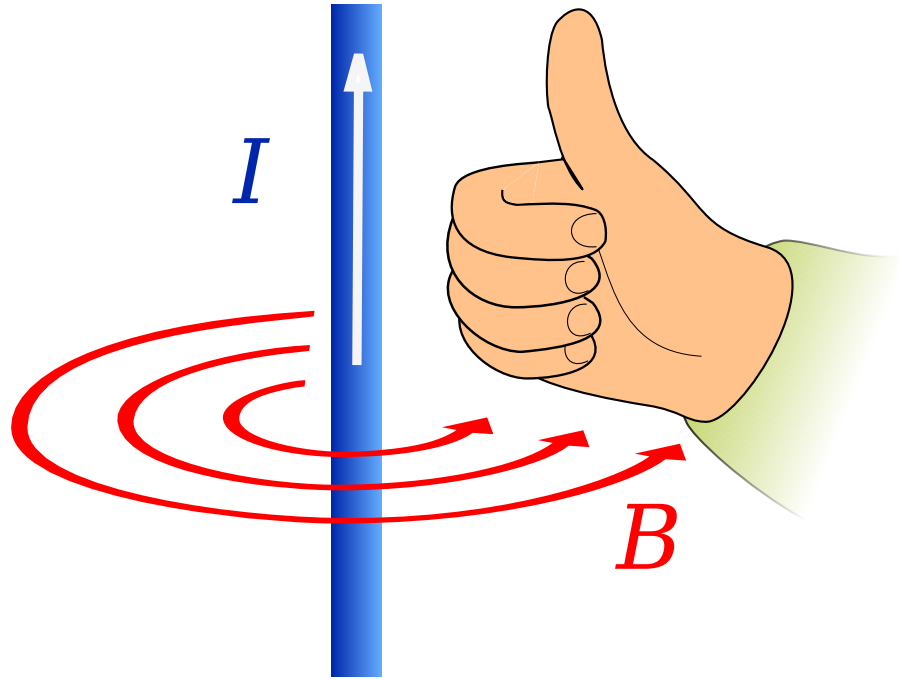
- Learning
 - what the ingredients of a forward model are
 - how the forward model links sources of the brain to sensors
 - what co-registration of MRI and MEG amounts to
 - how to fit dipoles
 - only reference material

Equivalent Current Dipole

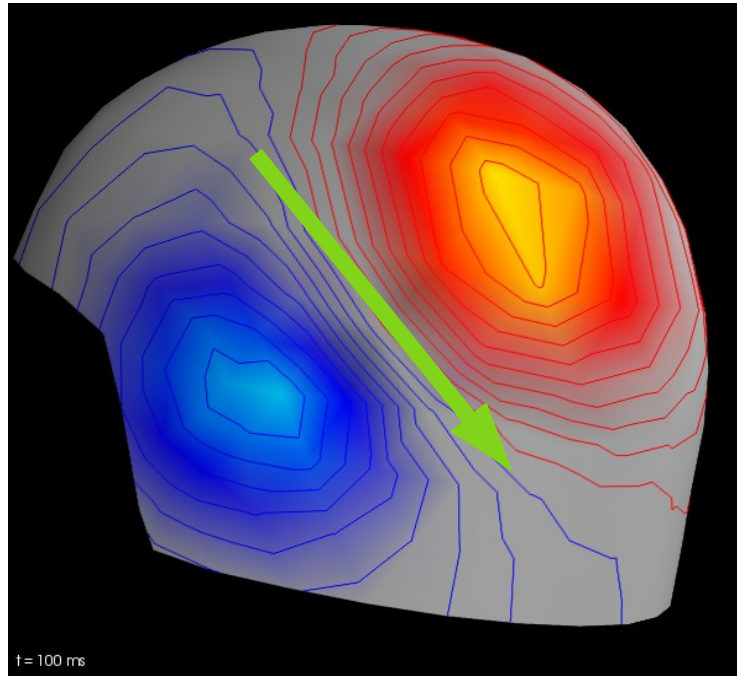


Stephanie Sillekens

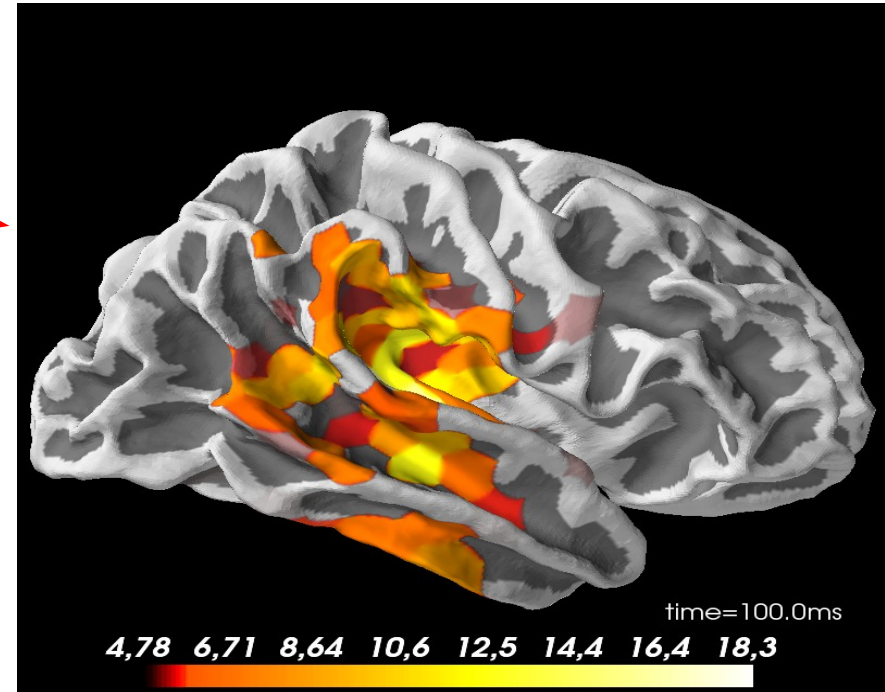
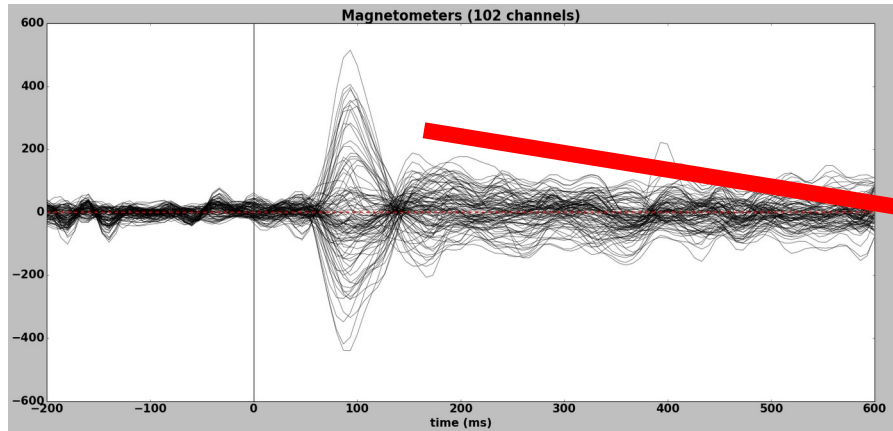
Right-hand rule



Field plot with very rough source localization



But... how do we get to here?

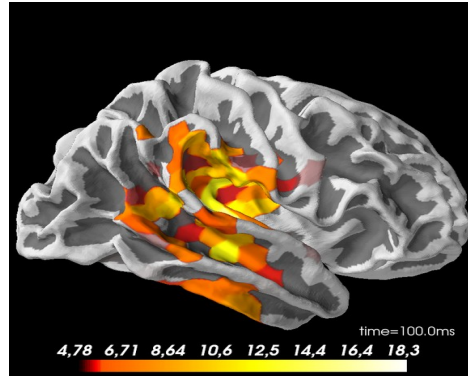


FORWARD MODELLING

(some slides from and inspired by FieldTrip)

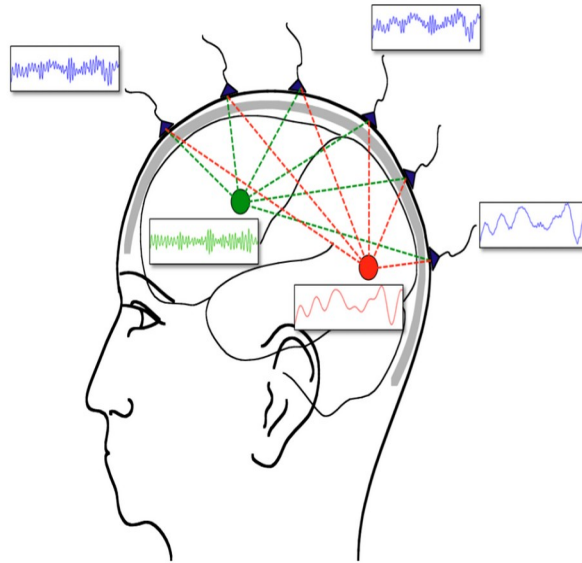
We want to go here:

Brain activation



Problem

Superposition of source activity



The problem in a nutshell

$$\mathbf{Y} = \mathbf{W}\mathbf{X}$$

Y: the true signal (magnetic field)

X: the neural generators (the underlying sources)

W: a weighting matrix (the leadfield)

The problem in a nutshell

$$\mathbf{b}(t) = \mathbf{L}(r) \mathbf{s}(r, t) + \mathbf{n}(t)$$

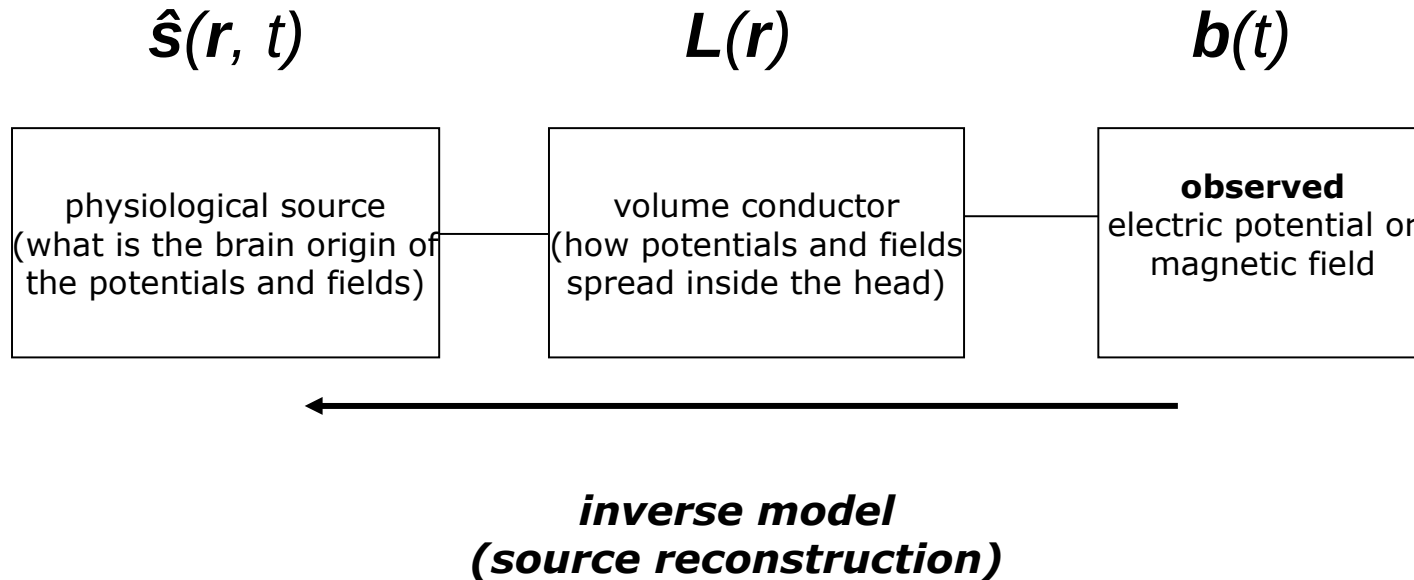
$\mathbf{b}(t)$: the measured magnetic field at time t

$\mathbf{s}(r, t)$: the source at position r at time t

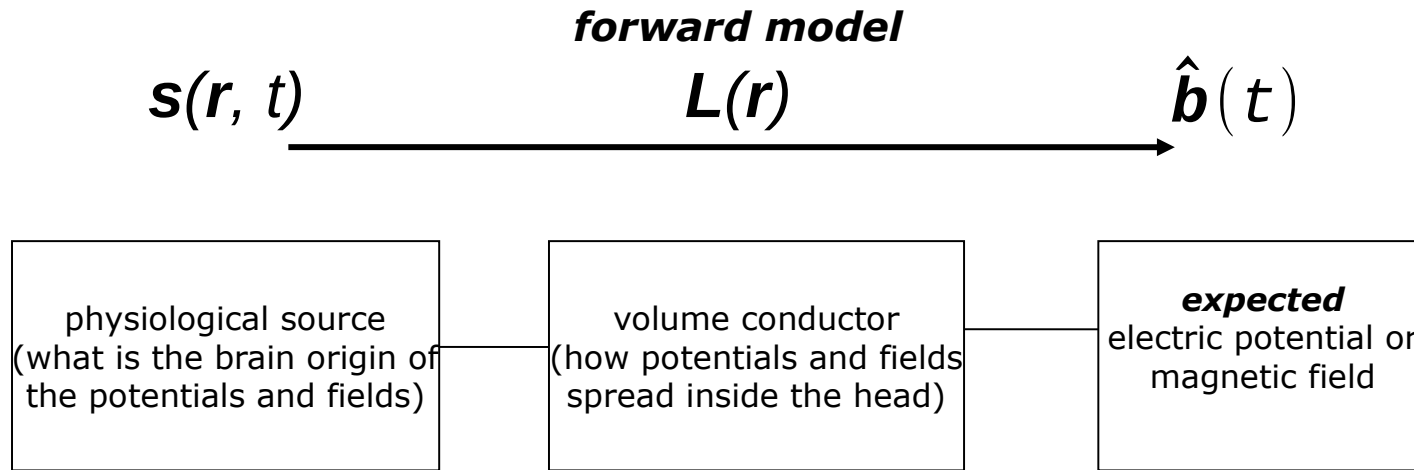
$\mathbf{L}(r)$: leadfield of source at position r (a weighting matrix)

$\mathbf{n}(t)$: normally distributed measurement noise at each time t

Inverse modelling



Forward modelling



IMPORTANT

Without a plausible forward model restricting the solution space, an infinite number of inverse models could explain the data

Ingredients for a forward model, $L(r)$

- A source model src
 - Telling us *the origin* of brain activity
- A volume conductor bem
 - Telling us how electric currents *spread* within the head bem solution
- Sensor positions info
 - Telling us *where* the sensors are relative to the sources
 - These need to be transformed into MR space

Info structure

```
In [29]: info = mne.io.read_info('workshop_2025_raw.fif')
```

```
In [30]: info['dev_head_t']
```

```
Out[30]:
```

```
<Transform | MEG device->head>
```

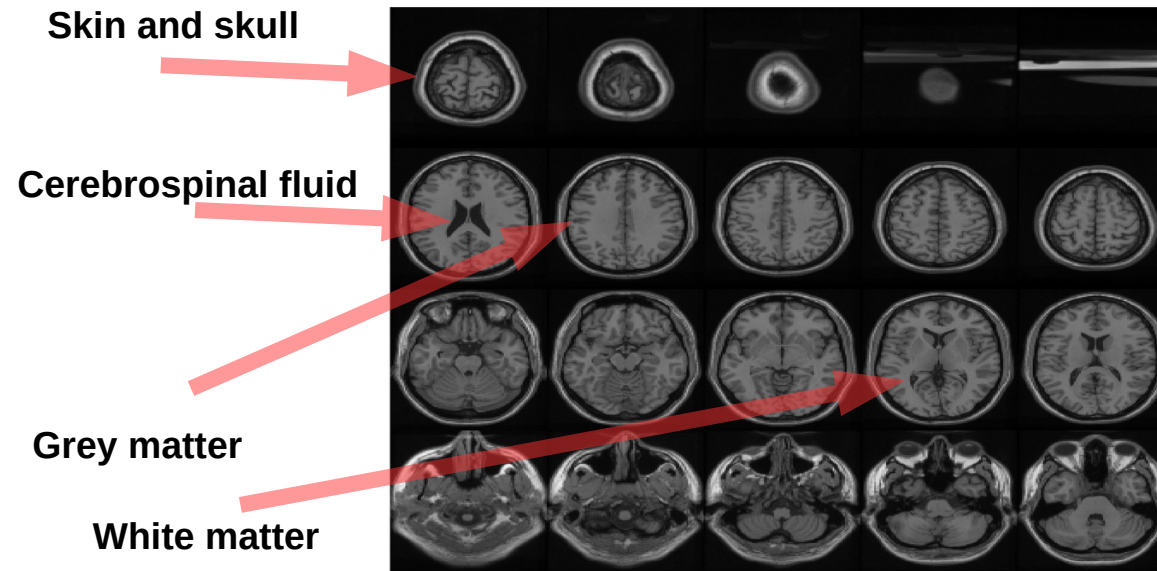
```
[[ 0.99853134  0.04444526  0.03098306 -0.00409153]
 [-0.04400658  0.99892318 -0.01469979  0.00523906]
 [-0.03160303  0.01331474  0.99941182  0.05708658]
 [ 0.          0.          0.          1.          ]]
```

```
In [32]: info['chs'][3]
```

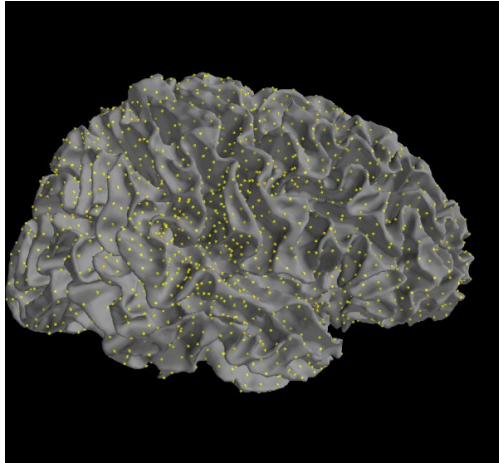
```
Out[32]:
```

```
{'scanno': 4,
 'logno': 111,
 'kind': 1 (FIFV_MEG_CH),
 'range': 1.9073486328125e-05,
 'cal': 1.329999999022391e-10,
 'coil_type': 3022 (FIFV_COIL_VV_MAG_T1),
 'loc': array([-0.1066      ,  0.0464      , -0.0604      , -0.0127      ,  0.0057      ,
               -0.99990302, -0.186801    , -0.98240298, -0.0033      , -0.98232698,
               0.18674099,  0.013541   ]),
 'unit': 112 (FIFV_UNIT_T),
 'unit_mul': 0 (FIFV_UNITM_NONE),
 'ch_name': 'MEG0111',
 'coord_frame': 1 (FIFV_COORD_DEVICE)}
```

What do we see in the MR?



Source model examples

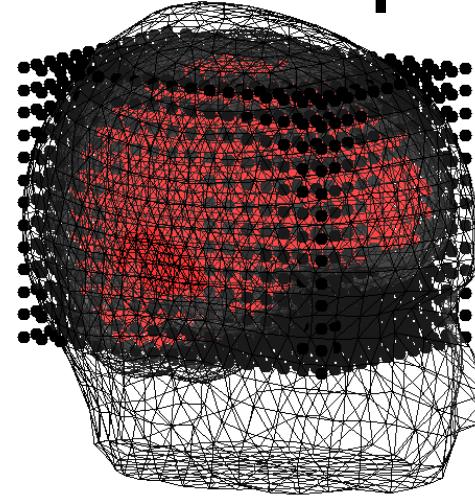


```
##% CORTICAL SURFACE SOURCE SPACE

for subject in subjects:
    cort_src = mne.source_space.setup_source_space(subject,
                                                    subjects_dir=subjects_dir,
                                                    n_jobs=-1)

    bem_path = join(subjects_dir, subject, 'bem')
    write_filename = subject + '-oct-6-src.fif'
    mne.source_space.write_source_spaces(join(bem_path, write_filename),
                                        cort_src)
```

Restricted to the cortical surface



```
##% VOLUME SOURCE SPACE

for subject in subjects:

    bem_path = join(subjects_dir, subject, 'bem')

    vol_src = mne.source_space.setup_volume_source_space(subject=subject,
                                                         bem=bem_solution,
                                                         subjects_dir=subjects_dir)

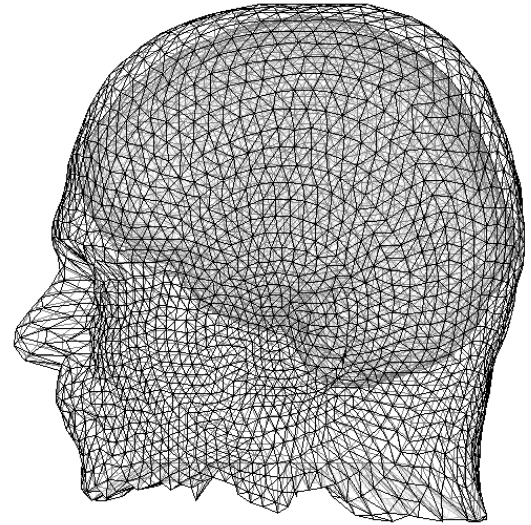
    write_filename = subject + '-volume-5-mm-src.fif'

    mne.source_space.write_source_spaces(join(bem_path, write_filename),
                                        vol_src)
```

Volumetric grid

Volume conductor (head model)

An anatomical model that models the conductivities of different tissues *bem*



Volume conductor (head model)

For *EEG*, we need to model the brain, skull and scalp with different conductivities (we are measuring volume currents)

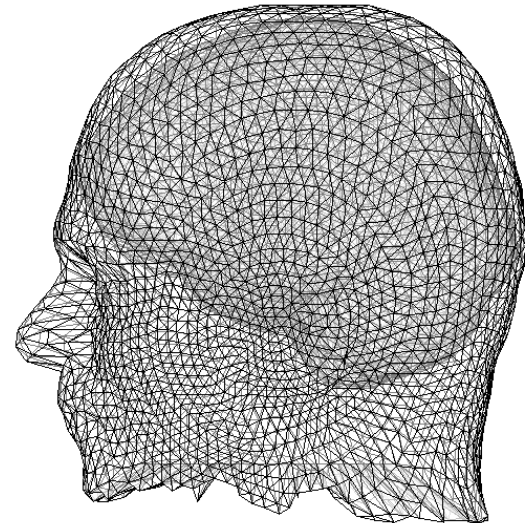
```
## BOUNDARY ELEMENT METHOD

## describe the surfaces and their conductivities
#bem_model = mne.bem.make_bem_model(subject=subject, subjects_dir=subjects_dir,
#                                   conductivity=[0.3, 0.006, 0.3]) ## three layer model

## model how electrical potentials spread to the electrodes and how the
# currents of the brain are related to the magnetic field measured at the
# sensors
#bem_solution = mne.bem.make_bem_solution(bem_model)

bem_solution = mne.bem.read_bem_solution('../subjects/sample/bem/sample-5120-5120-5120-bem-sol.fif')

print(bem_solution['solution'])
```



Volume conductor (head model)

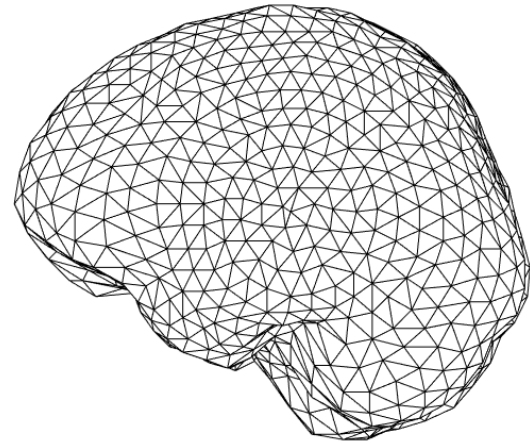
For MEG, the model can be much simpler since the magnetic field is roughly proportional to the magnitude of the primary current

(This is why MEG has better spatial resolution than EEG)

```
#!/usr/bin/env python
# %% BEM MODEL

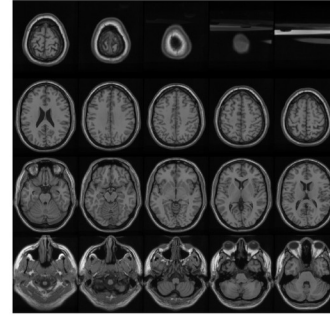
for subject in subjects:
    bem_path = join(subjects_dir, subject, 'bem')

    ## single-layer model
    write_filename = subject + '-5120-bem.fif'
    bem_surfaces = mne.bem.make_bem_model(subject, conductivity=[0.3],
                                          subjects_dir=subjects_dir)
    mne.bem.write_bem_surfaces(join(bem_path, write_filename),
                              bem_surfaces)
```

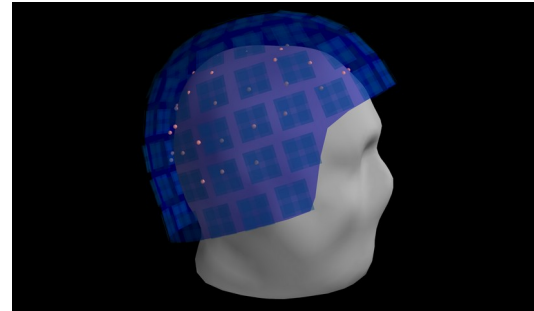


Sensor positions

- The MR-images and the helmet/electrode data will be represented in different coordinate systems
 - Thus, they have to be co-registered



≠



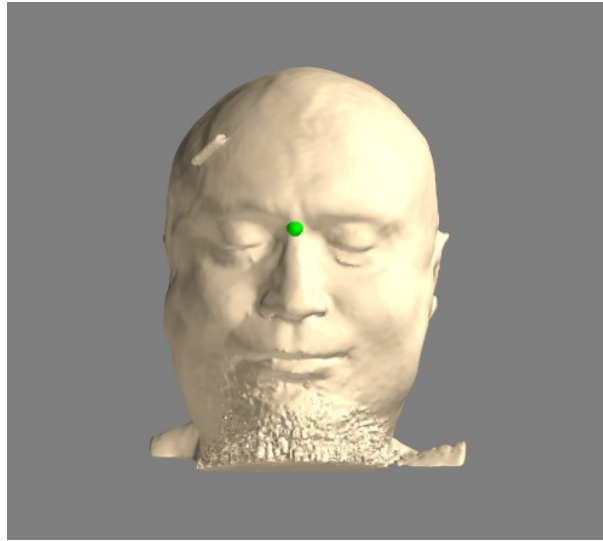
Co-registration



Construct a head model based on the MRI

Co-registration

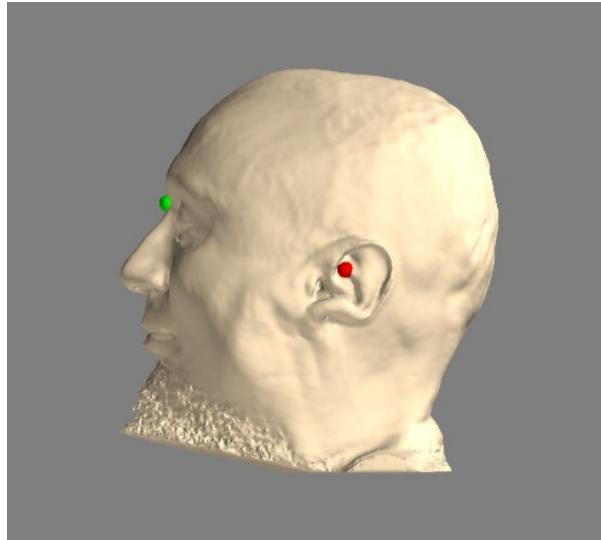
1. Nasion



Plot in fiducial points (these are acquired before recording)

Co-registration

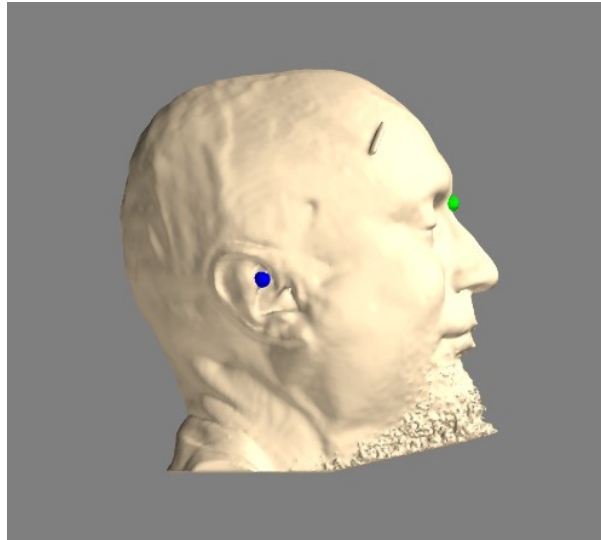
1. Nasion
2. Left pre-auricular point



Plot in fiducial points (these are acquired before recording)

Co-registration

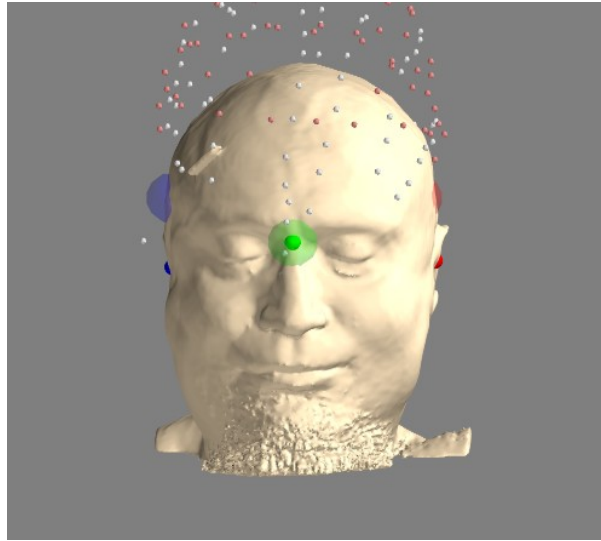
1. Nasion
2. Left pre-auricular point
3. Right pre-auricular point



Plot in fiducial points (these are acquired before recording)

Co-registration

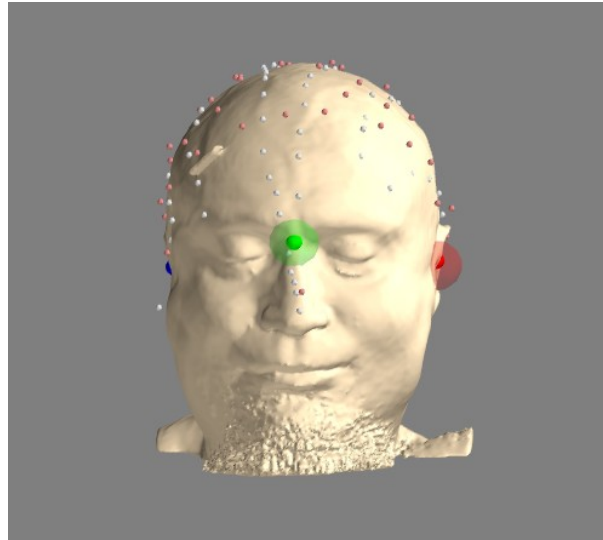
1. Nasion
2. Left pre-auricular point
3. Right pre-auricular point



Plot extra head points

Co-registration

1. Nasion
2. Left pre-auricular point
3. Right pre-auricular point

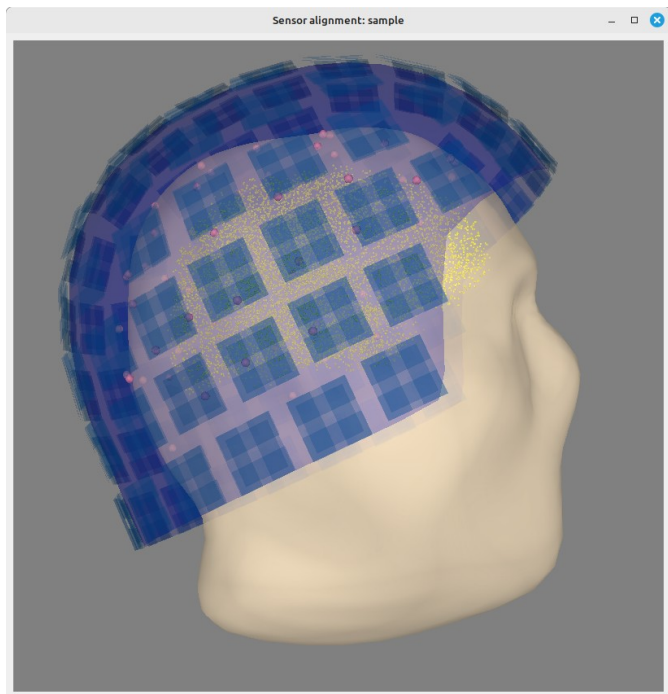


```
In [7]: trans = mne.read_trans('workshop_2025-trans.fif')
In [8]: trans
Out[8]:
<Transform | head->MRI (surface RAS)>
[[ 0.99968028 -0.02514233 -0.00269039  0.00789862]
 [ 0.02285611  0.94399983 -0.3291533  -0.00342108]
 [ 0.01081541  0.32898656  0.9442727  -0.03508078]
 [ 0.         0.         0.         1.         ]]
```

```
In [32]: info['chs'][3]
Out[32]:
{'scanno': 4,
 'logno': 111,
 'kind': 1 (FIFV_MEG_CH),
 'range': 1.9073486328125e-05,
 'cal': 1.329999999022391e-10,
 'coil_type': 3022 (FIFV_COIL_VV_MAG_T1),
 'loc': array([-0.1066,  0.0464, -0.0604, -0.0127,  0.0057,
               -0.99990302, -0.186801, -0.98240298, -0.0033, -0.98232698,
               0.18674099,  0.013541 ]),
 'unit': 112 (FIFV_UNIT_T),
 'unit_mul': 0 (FIFV_UNITM_NONE),
 'ch_name': 'MEG0111',
 'coord_frame': 1 (FIFV_COORD_DEVICE)}
```

Use an algorithm to minimize the distance between points and head shape. The MR and the MEG are now co-registered

Check alignment



```
# extract source space
src = fwd['src']
##plot alignment does not work on UCloud
mne.viz.plot_alignment(info=epochs_sample.info, src=src,
                      subjects_dir='../..../subjects/',
                      subject="sample", surfaces="outer_skin",
                      trans='sample_audvis_raw-trans.fif')
```

With all ingredients in place

WE CAN GET THE FORWARD MODEL

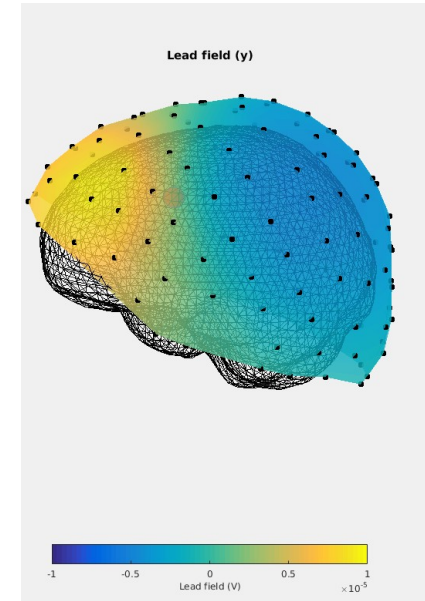
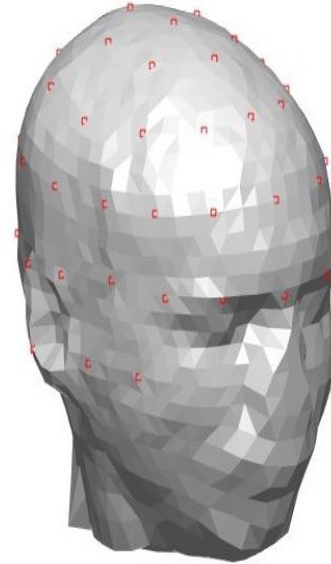
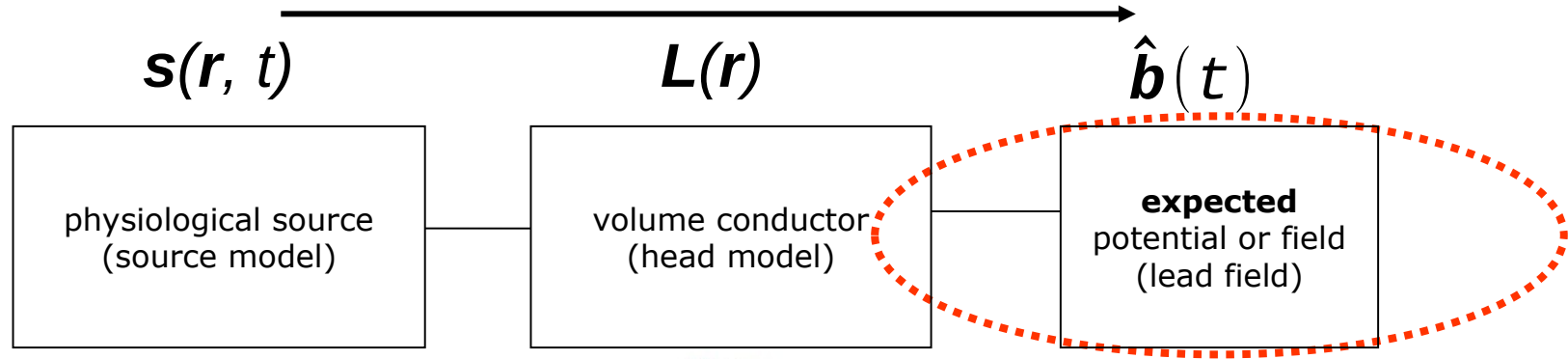
- Volume conduction model
 - *bem*
- Source model
 - *src*
- Transformed sensor positions
 - *info*
 - *trans*

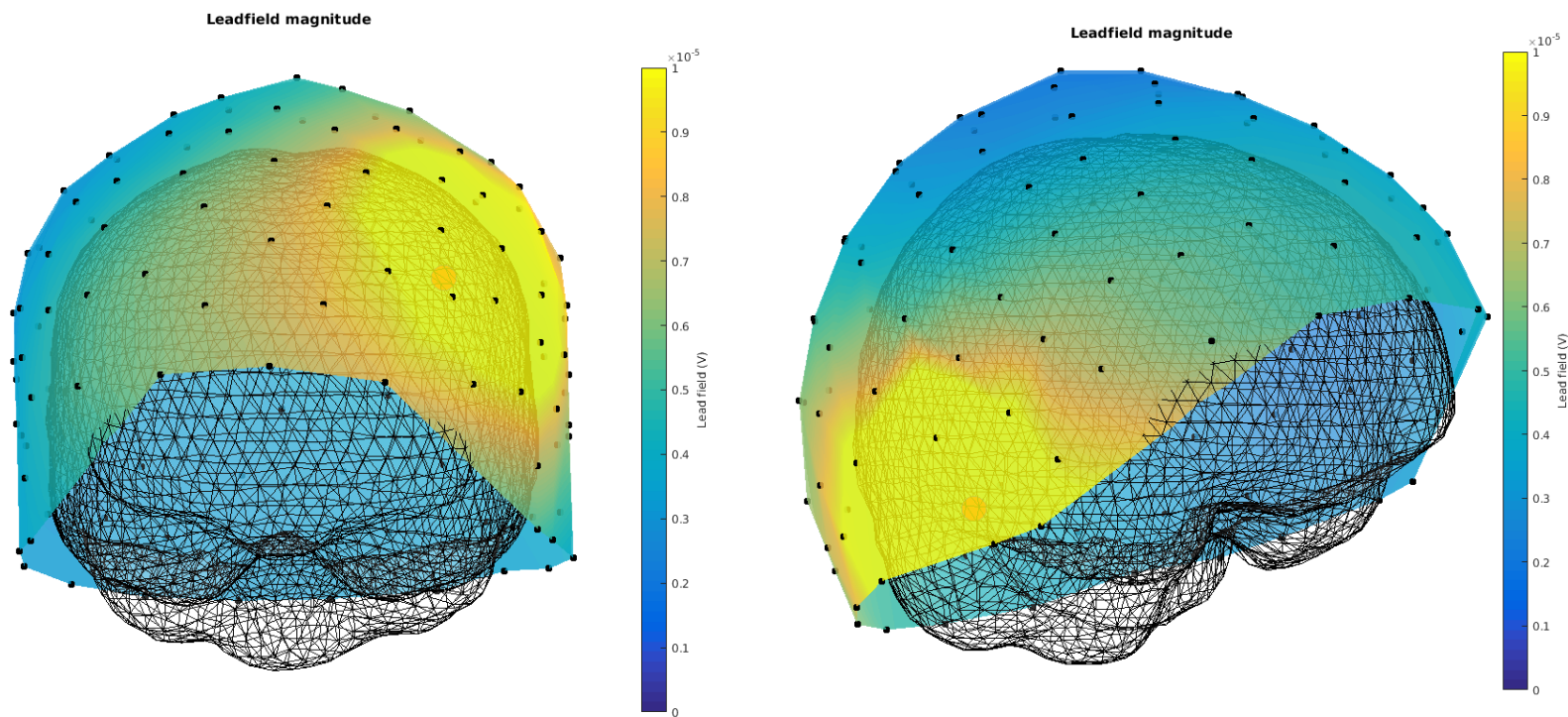
```
## GET FORWARD MODEL

info = evoked.info
trans = join(MEG_path, ?subject? + '-trans.fif')
src = join(subjects_dir, ?subject?, 'bem', ?subject? + '-oct-6-src.fif')
bem = join(subjects_dir, ?subject?, 'bem', ?subject? + '-5120-bem-sol.fif')

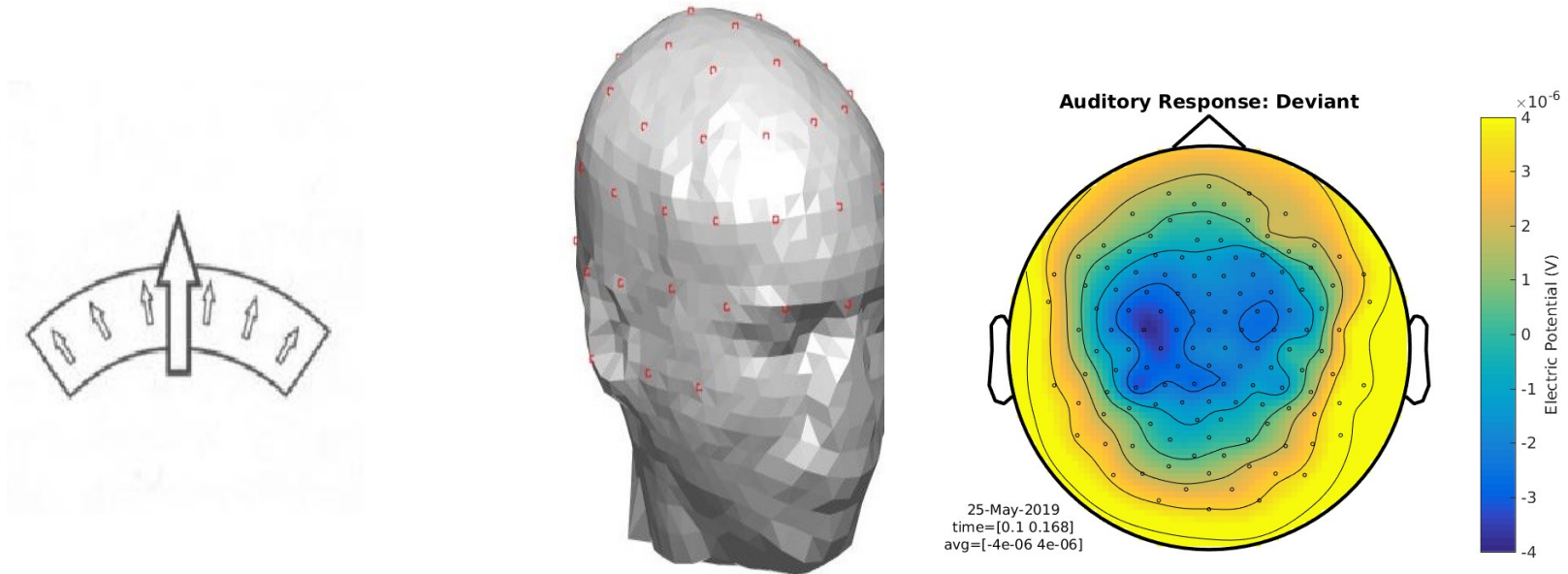
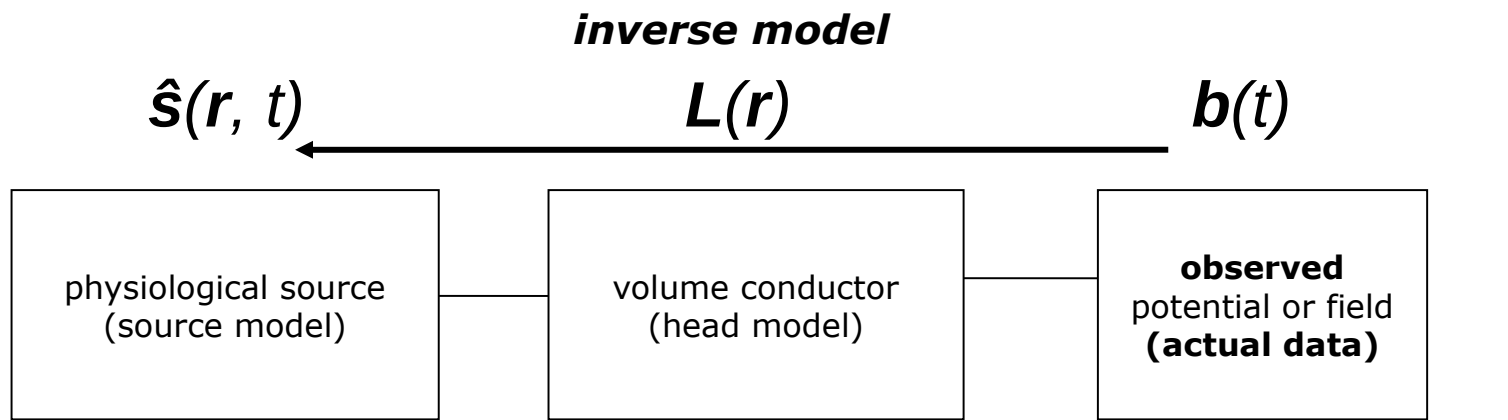
fwd = mne.make_forward_solution(
    info, trans,
    src, bem)
```

forward model





and so on for each source in the source model ...



Now we can create the forward model (leadfield)

The forward model models how each source in the source space *would* be seen by the sensors (in T) in the helmet, *given* that that source is active with 1 Am

In other words; the sensitivity of MEG to different sources

Units

$$\mathbf{b}(t) = \mathbf{L}(r) \mathbf{s}(r, t) + \mathbf{n}(t)$$

$$[T] = \left[\frac{T}{Am} \right] [Am] + [T]$$

Contents of the *fwd*

WHAT IS THE STORY I WANT TO TELL HERE?

```
In [44]: fwd['sol']['data'].shape  
Out[44]: (366, 22494)
```

```
In [49]: len(evoked_sample.ch_names)  
Out[49]: 366
```

```
In [56]: fwd['sol']['data'].shape[1] // 3  
Out[56]: 7498
```

```
In [58]: evoked_sample.ch_names[2]  
Out[58]: 'MEG 0111'
```

```
In [50]: fwd['src']  
Out[50]: <SourceSpaces: [<surface (lh), n_vertices=155407, n_used=3732>,  
<surface (rh), n_vertices=156866, n_used=3766>] head coords, subject  
'sample', ~31.0 MiB>
```

```
In [51]: len(fwd['src'])  
Out[51]: 2
```

```
In [52]: fwd['src'][0]['nuse'] + fwd['src'][1]['nuse']  
Out[52]: 7498
```

```
In [57]: fwd['sol']['data'][2, :]  
Out[57]:  
array([-7.1391241e-07,  9.5892676e-07,  7.8583281e-07, ...,  
        9.4759559e-07, -2.4599572e-07, -6.7779990e-07],  
      shape=(22494,), dtype=float32)
```

What order will the magnetic field (in T) roughly be, when we have sources with current densities of 10-100 nAm ?

```
In [57]: fwd['sol']['data'][2, :]  
Out[57]:  
array([-7.1391241e-07,  9.5892676e-07,  7.8583281e-07, ...,  
        9.4759559e-07, -2.4599572e-07, -6.7779990e-07],  
       shape=(22494,), dtype=float32)
```

Interim summary

- The forward model consists of:
 - a source space (*src*)
 - a solution of the spread of the currents (*bem*)
 - transformed (*trans*) positions of the sensor positions (*info*)
- The forward model links the source space (*src*) to the transformed sensor space (*trans*info*) through the model (*bem*)
 - Its unit is: $\left[\frac{T}{Am} \right]$

Learning goals

ONLY FOR REFERENCE

- For the dipole modelling
 - understanding when it is appropriate to use dipole fitting, minimum norm estimate and beamformer strategies respectively
 - understanding how the forward model is a crucial ingredient in all source modelling strategies
 - get a first understanding of the relevant equations

Inverse model types

- Dipole fitting
 - Only a single or a few dipolar sources are active at given time sample
 - The solution is *overdetermined*: there are more data channels (hundreds) than sources (a few)
- Minimum Norm Estimate
 - All (cortical) sources are active all the time
 - The solution is *underdetermined*: there are more sources (thousands) than channels (hundreds)
- Beamformer
 - All sources are estimated independently from one another

$$[\mathbf{b}(t) - \mathbf{L}(\mathbf{r})\mathbf{L}(\mathbf{r})^+ \mathbf{b}(t)]^2$$

$$\hat{\mathbf{v}}_{\text{vox}}(t) = \mathbf{L}_V^T (\mathbf{G} + \epsilon \mathbf{I})^{(-1)} \mathbf{b}(t)$$

$$\mathbf{w}(\mathbf{r}) = \frac{\mathbf{R}^{-1} \mathbf{L}(\mathbf{r})}{\mathbf{L}^T(\mathbf{r}) \mathbf{R}^{-1} \mathbf{L}(\mathbf{r})}$$

Inverse model types

- Dipole fitting
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$$w(\mathbf{r}) = \frac{\mathbf{R}^{-1} \mathbf{L}(\mathbf{r})}{\mathbf{L}^T(\mathbf{r}) \mathbf{R}^{-1} \mathbf{L}(\mathbf{r})}$$

Underdetermined: more unknowns, n , than knowns, m

Overdetermined: more knowns, m , than unknowns, n

Learning goals

- Learning
 - how to derive the minimum-norm estimate
 - how the minimum-norm estimate forces all activity in $\mathbf{b}(t)$ to go into $\hat{\boldsymbol{\nu}}_{vox}(t)$
 - Why dSPM is the default choice?

A solution to our basic equation

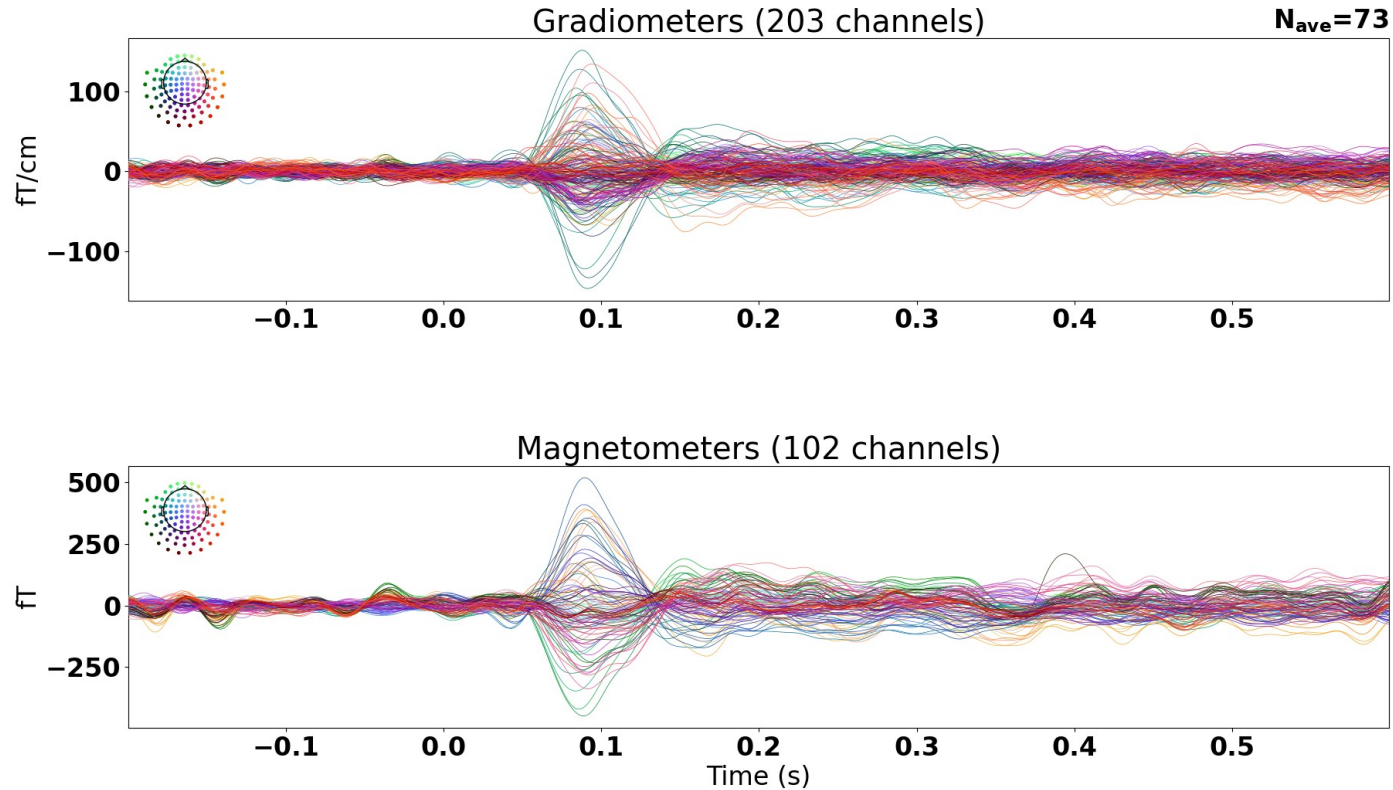
$$\mathbf{b}(t) = \mathbf{L}(\mathbf{r})\mathbf{s}(\mathbf{r}, t) + \mathbf{n}(t)$$

$$\mathbf{b}(t) = \mathbf{L}_V \boldsymbol{\nu}_{vox}(t) + \mathbf{n}(t)$$

$$\hat{\boldsymbol{\nu}}_{vox}(t) = \mathbf{L}_V^T (\mathbf{G} + \epsilon \mathbf{I})^{-1} \mathbf{b}(t)$$

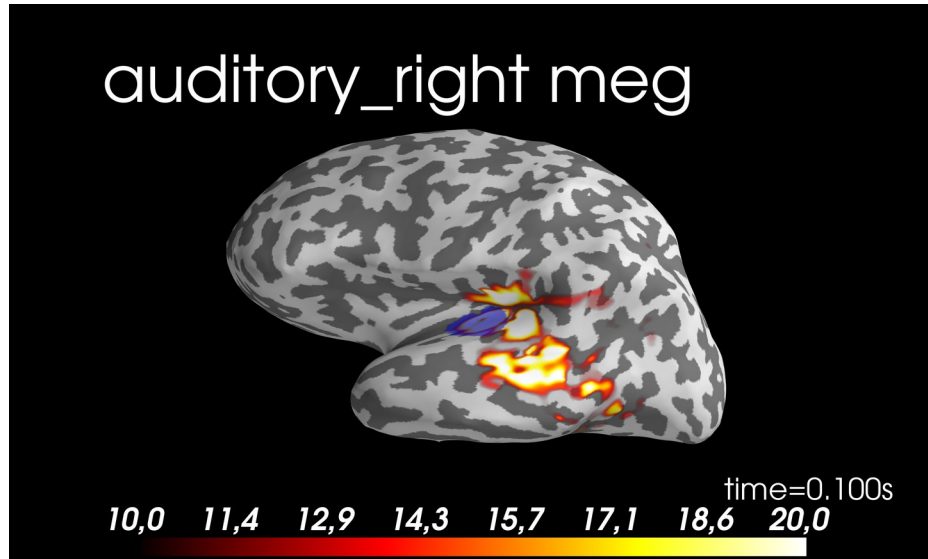
MNE – visual example

Auditory responses (right ear)

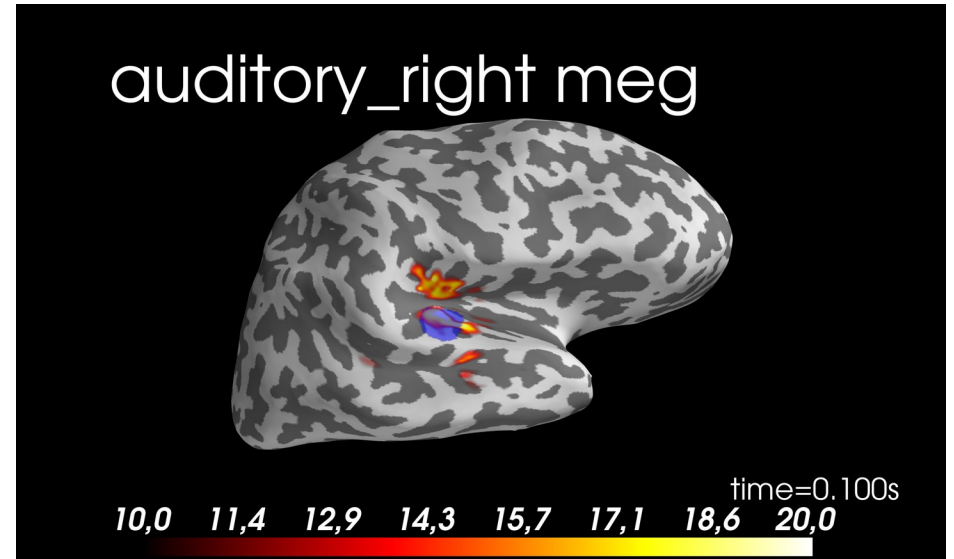


Minimum norm estimate

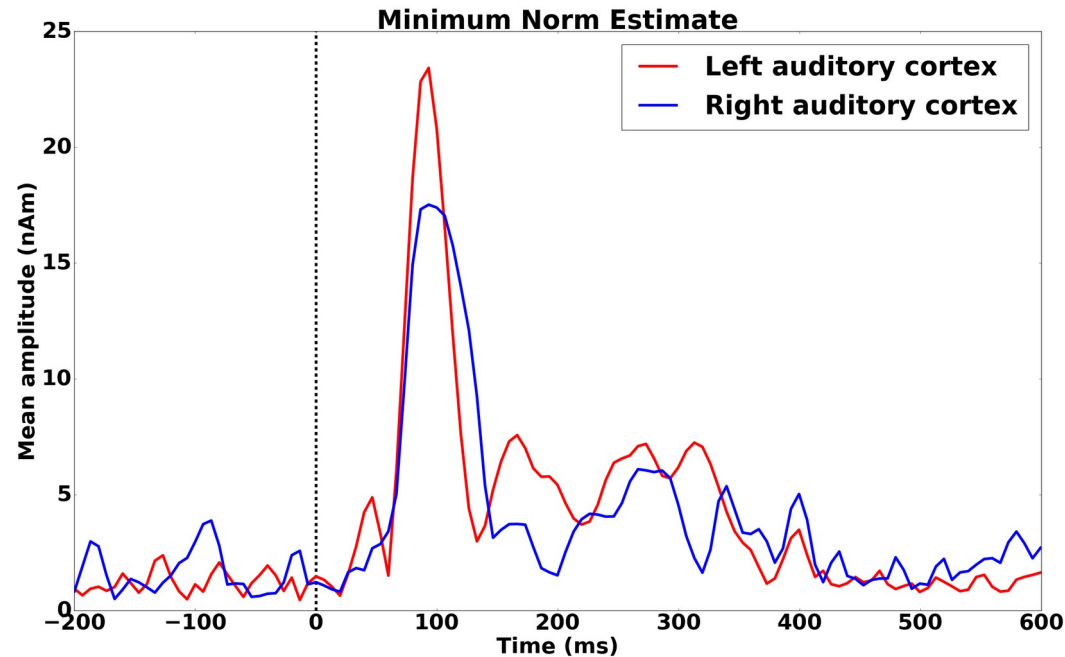
Left hemisphere



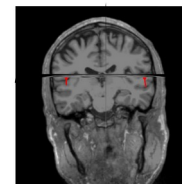
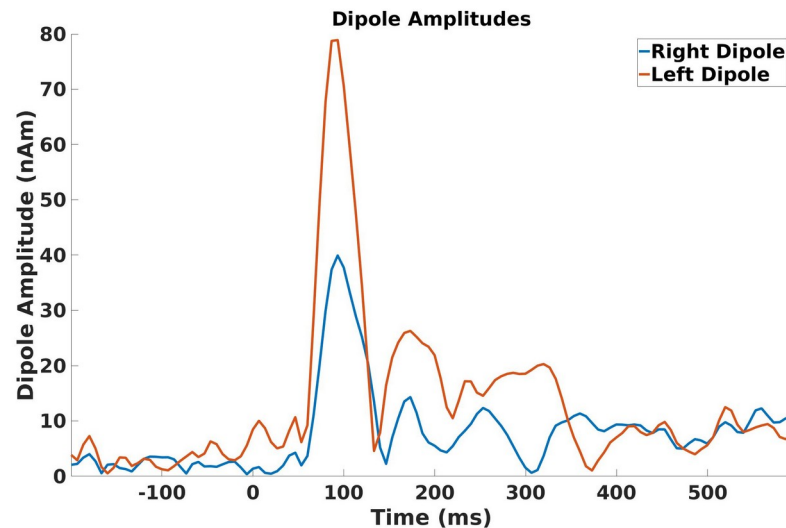
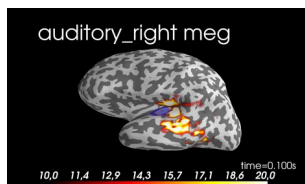
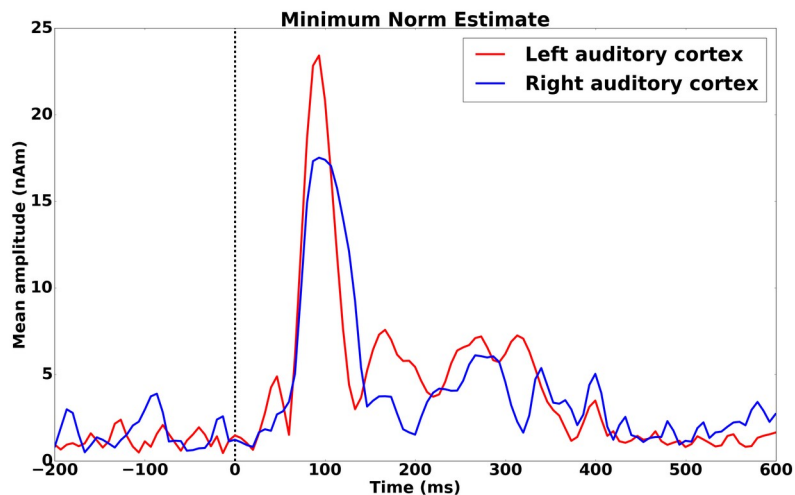
Right hemisphere

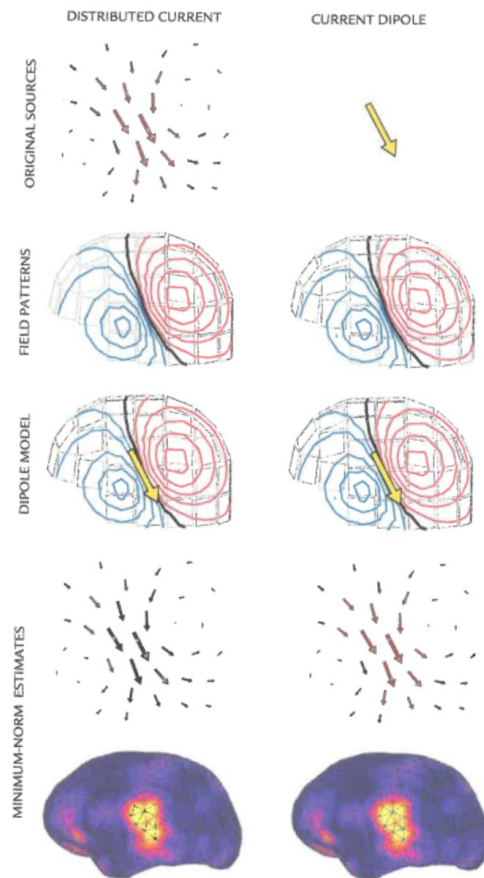


Minimum norm estimate – time courses



Comparison between dipole and minimum norm estimates





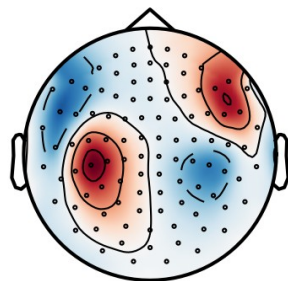
Hari and Puce,
2017

Comparisons between dipole fits and MNE

- Dipole fit

- Very good for modelling early sensory responses that have a dipolar topography

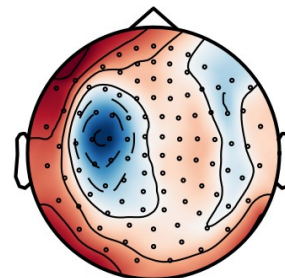
100 ms



- MNE

- Can model more complex responses than dipole fits

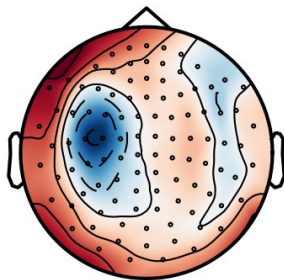
300 ms



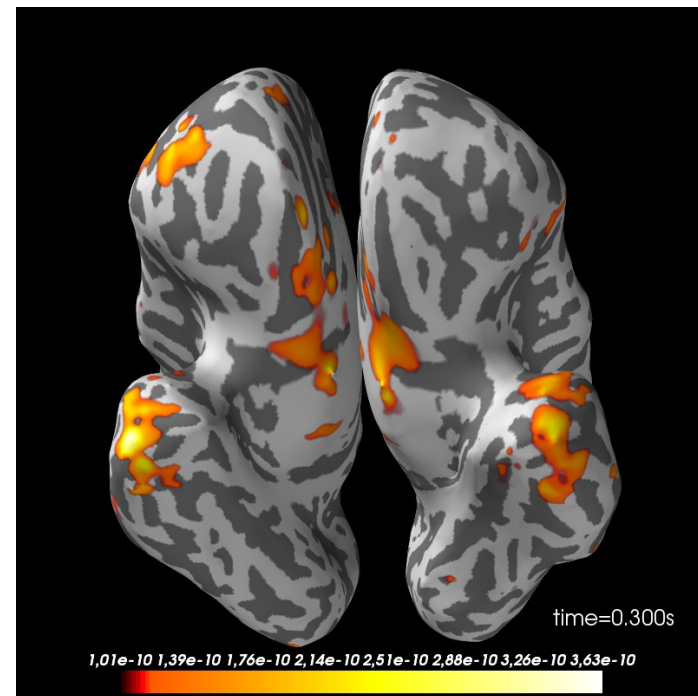
Comparisons between dipole fits and MNE

- MNE
 - Can model more complex responses than dipole fits

300 ms



SEEMS to be
a more
temporal (lobe)
response



Your job today – get from 2nd to 3rd
equation

$$\mathbf{b}(t) = \mathbf{L}(\mathbf{r})\mathbf{s}(\mathbf{r}, t) + \mathbf{n}(t)$$

$$\mathbf{b}(t) = \mathbf{L}_V \boldsymbol{\nu}_{vox}(t) + \mathbf{n}(t)$$

$$\hat{\boldsymbol{\nu}}_{vox}(t) = \mathbf{L}_V^T (\mathbf{G} + \epsilon \mathbf{I})^{-1} \mathbf{b}(t)$$

$$\begin{aligned}
\mathbf{b}(t) &= \begin{bmatrix} b_1(t) \\ b_2(t) \\ \vdots \\ b_M(t) \end{bmatrix} & \mathbf{L}(\mathbf{r}) &= [\mathbf{l}_x(\mathbf{r}), \mathbf{l}_y(\mathbf{r}), \mathbf{l}_z(\mathbf{r})] \\
& & \mathbf{n}(t) &= \begin{bmatrix} n_1(t) \\ n_2(t) \\ \vdots \\ n_M(t) \end{bmatrix} & \boldsymbol{\nu}_{vox}(t) &= \begin{bmatrix} s(\mathbf{r}_1, t) \\ s(\mathbf{r}_2, t) \\ \vdots \\ s(\mathbf{r}_N, t) \end{bmatrix} \\
\mathbf{G} &\approx \mathbf{L}_V \mathbf{L}_V^T & \mathbf{s}(\mathbf{r}, t) &= \begin{bmatrix} s_x(\mathbf{r}, t) \\ s_y(\mathbf{r}, t) \\ s_z(\mathbf{r}, t) \end{bmatrix} \\
\mathbf{L}_V &= [\mathbf{L}(\mathbf{r}_1), \mathbf{L}(\mathbf{r}_2), \dots, \mathbf{L}(\mathbf{r}_N)]
\end{aligned}$$

Restrictions

- No computers
- No books
- Only pieces of paper
- Do this(roughly) in your study groups

I will provide help along the way

(You can ask me about generalised inverses if you need that at one point)

Tip, write the dimensions of the matrices below to check that the multiplications are possible (inner dimensions have to agree)

$$\mathbf{A}_{a \times b} \mathbf{B}_{b \times a} = \mathbf{C}_{a \times a}$$

$$\mathbf{A}_{a \times b} \mathbf{B}_{a \times b}: \text{not defined}$$

When you have derived the MNE, try telling the story the same way I did for the forward model based on your data
(You can open your computers now)

Contents of the *fwd*

STORY I TOLD LAST TIME

```
In [44]: fwd['sol']['data'].shape  
Out[44]: (366, 22494)
```

```
In [49]: len(evoked_sample.ch_names)  
Out[49]: 366
```

```
In [56]: fwd['sol']['data'].shape[1] // 3  
Out[56]: 7498
```

```
In [58]: evoked_sample.ch_names[2]  
Out[58]: 'MEG 0111'
```

```
In [50]: fwd['src']  
Out[50]: <SourceSpaces: [<surface (lh), n_vertices=155407, n_used=3732>,  
<surface (rh), n_vertices=156866, n_used=3766>] head coords, subject  
'sample', ~31.0 MiB>
```

```
In [51]: len(fwd['src'])  
Out[51]: 2
```

```
In [52]: fwd['src'][0]['nuse'] + fwd['src'][1]['nuse']  
Out[52]: 7498
```

```
In [57]: fwd['sol']['data'][2, :]  
Out[57]:  
array([-7.1391241e-07,  9.5892676e-07,  7.8583281e-07, ...,  
        9.4759559e-07, -2.4599572e-07, -6.7779990e-07],  
      shape=(22494,), dtype=float32)
```

Contents of the *MNE*, *fwd* and *evoked*

TELL YOUR STORY HERE

Cost function $\mathcal{F}$

$$\mathcal{F} = ||\mathbf{b}(t) - \mathbf{L}_V \hat{\mathbf{v}}_{vox}(t)||^2 + \epsilon ||\hat{\mathbf{v}}_{vox}(t)||^2$$

is minimised by

$$\hat{\mathbf{v}}_{vox}(t) = \mathbf{L}_V^T (\mathbf{G} + \epsilon \mathbf{I})^{-1} \mathbf{b}(t)$$

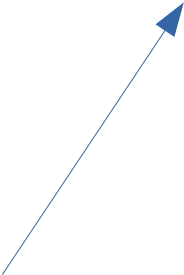
Voila – the minimum norm estimate

– optimising two constraints

$$\mathcal{F} = ||\mathbf{b}(t) - \mathbf{L}_V \hat{\mathbf{v}}_{vox}(t)||^2 + \epsilon ||\hat{\mathbf{v}}_{vox}(t)||^2$$



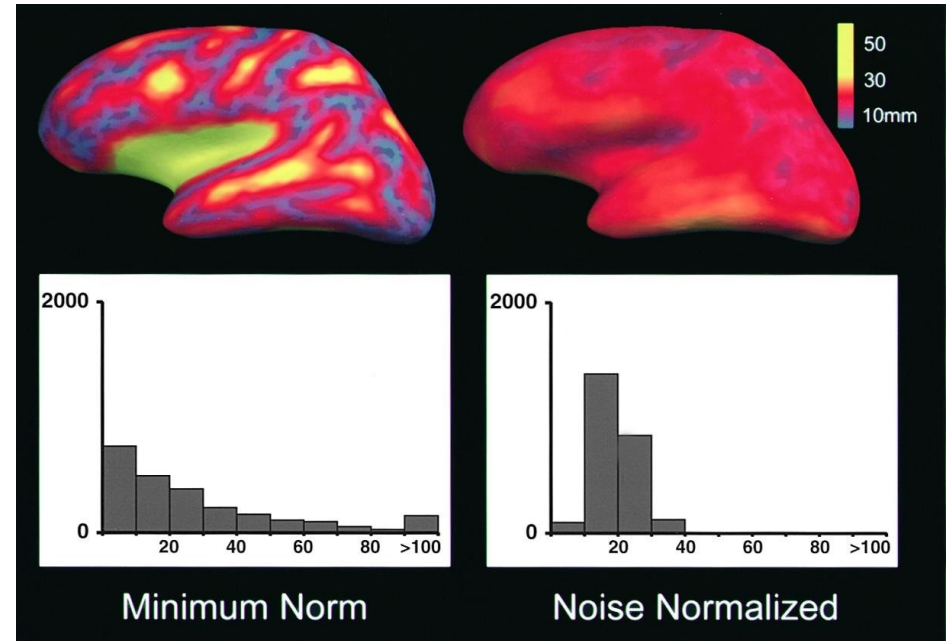
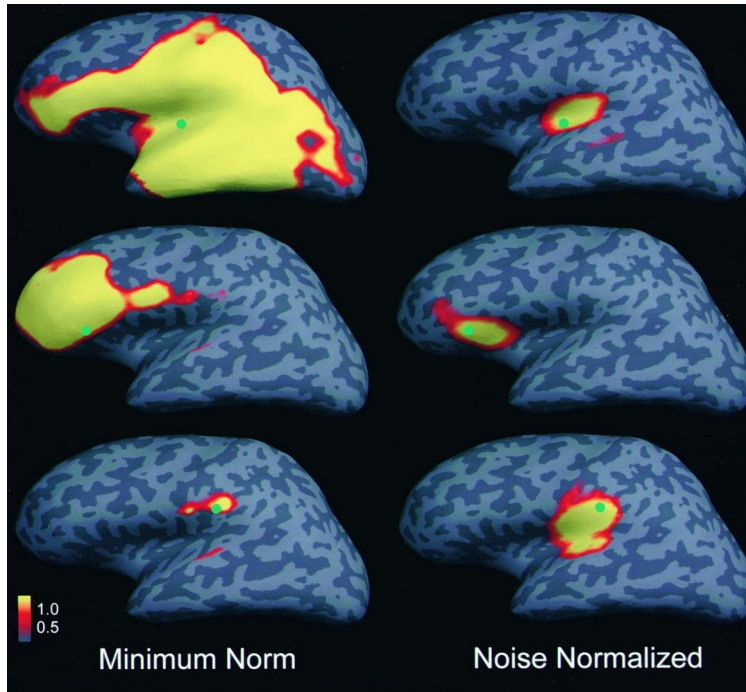
Minimising the
unexplained variance of
the solution



Minimising the norm of
the solution (less
current is better than
more current)

Surface bias

Mitigated by choosing the dSPM method



Dale AM, Liu AK, Fischl BR, et al (2000) Dynamic Statistical Parametric Mapping: Combining fMRI and MEG for High-Resolution Imaging of Cortical Activity. *Neuron* 26:55–67.
[https://doi.org/10.1016/S0896-6273\(00\)81138-1](https://doi.org/10.1016/S0896-6273(00)81138-1)

dSPM

mne.minimum_norm.apply_inverse

```
mne.minimum_norm.apply_inverse(evoked, inverse_operator,  
lambda2=0.1111111111111111, method='dSPM', pick_ori=None, prepared=False,  
label=None, method_params=None, return_residual=False, use_cps=True,  
verbose=None)
```

[\[source\]](#)

Apply inverse operator to evoked data.

Parameters:

evoked : [Evoked](#) object

Evoked data.

inverse_operator : instance of [InverseOperator](#)

Inverse operator.

lambda2 : [float](#)

The regularization parameter.

method : "MNE" / "dSPM" / "sLORETA" / "eLORETA"

Use minimum norm [\[1\]](#), dSPM (default) [\[2\]](#), sLORETA [\[3\]](#), or eLORETA [\[4\]](#).

Summary

- The minimum-norm estimate explains the whole field, $\mathbf{b}(t)$ and uses all defined sources $\hat{\mathbf{v}}_{vox}(t)$
 - thus, $\mathbf{b}(t)$ (preferably) has to be clean from all non-brain related activity
- It does by balancing explaining variance, and using the least amount of energy doing so.
 - means it will be biased to the surface
 - the dSPM will mitigate this bias

The course plan

Week 37:

Monday: Workshop paradigm: Measuring visual subjective experience
MR Recordings (Lau)

Week 38:

Monday: What is it all about? (Lau)
Workshop 1: MR-recordings (Laura & Lau)

Week 39:

Monday: Basic physiology and Evoked responses (Lau)
Wednesday: Preprocessing and evoked responses (Laura)

Week 40:

Monday: Forward modelling and inverse modelling (Lau)
Wednesday: Source reconstruction (Laura)

Week 41:

Monday: Multivariate statistics (Laura) (Lecture)
Wednesday: Multivariate statistics (Laura) (Coding)
Deadline for feedback: Video Explainer

Week 43:

Monday: Inverse modelling: beamforming (Lau) (Lecture)
Wednesday: Inverse modelling: beamforming (Laura) (Coding)

Week 44:

Monday From single subject analysis to group analysis (Laura)
Wednesday: Work on your reports (Laura)

Week 45:

Monday: What about that other cortex? - the cerebellar one (Lau)
Wednesday: Work on your reports, continued (Laura)

Week 46:

Monday: Optically pumped magnetoencephalography (Laura)
Wednesday: Respiration (Laura)
Deadline for feedback: Lab report

Week 47:

No teaching: prepare presentations!

Week 48:

Monday: Oral presentations (part 1) (Lau)
Wednesday: Oral presentations (part 2) (Lau)

Week 49:

Monday: Oral presentations (part 3) (Lau)
Wednesday: Oral presentations (part 4) (Lau)

Week 50:

Wednesday: What was it all about (Lau)

Reading questions (week 43)

- Focus on sections 4.1, 4.4 and 4.7
 - And have a good look at figures 4.4, 4.5 and 4.6
- What does it mean that a filter is spatial?
 - And what does it mean that it is adaptive?

Minimum Norm Estimate

$$\hat{\mathbf{v}}_{vox}(t) = \mathbf{L}_V^T (\mathbf{G} + \epsilon \mathbf{I})^{-1} \mathbf{b}(t).$$

$\hat{\mathbf{v}}_{vox}(t)$ = all the sources in the brain at a given time point

$\mathbf{L}_V^T$ = the leadfield (the estimated forward model)

$\mathbf{G}$ = the Gram matrix ($\mathbf{G} \approx \mathbf{L}_V \mathbf{L}_V^T$)

ϵ = a constant

$\mathbf{I}$ = the identity matrix

$\mathbf{b}(t)$ = the observed magnetic field or the electric potential at a given time point

We thus only need the leadfield and the observed data to estimate source activity

$(\hat{\mathbf{v}}_{vox}(t))$

Solution for deriving the MNE

Equations

$$\mathbf{b}(t) = \mathbf{L}_v \mathbf{v}_{vox}(t)$$

if these were scalar variables,
how would you isolate $\mathbf{v}_{vox}(t)$
(ignoring $\mathbf{n}(t)$?

We cannot divide with matrices, but multiplying by the inverse is possible, e.g.

$$\mathbf{b}(t) = \mathbf{L}_v \mathbf{v}_{vox}(t)$$

$$\mathbf{L}_v^{-1} \mathbf{b}(t) = \mathbf{L}_v^{-1} \mathbf{L}_v \mathbf{v}_{vox}(t)$$

$$\mathbf{L}_v^{-1} \mathbf{b}(t) = \mathbf{I} \mathbf{v}_{vox}(t)$$

$$\mathbf{L}_v^{-1} \mathbf{b}(t) = \mathbf{v}_{vox}(t)$$

$$\mathbf{v}_{vox}(t) = \mathbf{L}_v^{-1} \mathbf{b}(t)$$

but $\mathbf{L}_v$ is an $M \times 3N$ matrix and $M < 3N$

(M = number of sensors, N = number of sources)

Is the inverse matrix
of $\mathbf{L}_v$ then defined?

No, but the generalised inverse is $\mathbf{L}_v^+$

$$\mathbf{L}_v^+ = \mathbf{L}_v^T [\mathbf{L}_v \mathbf{L}_v^T]^{(-1)}$$

thus we can isolate $\mathbf{v}_{vox}(t)$ by multiplying both sides with the generalised inverse

$$\mathbf{b}(t) = \mathbf{L}_v \mathbf{v}_{vox}(t)$$

and our estimate for $\mathbf{v}_{vox}(t)$ becomes:

$$\hat{\mathbf{v}}_{vox}(t) = \mathbf{L}_v^T [\mathbf{L}_v \mathbf{L}_v^T]^{(-1)} \mathbf{b}(t)$$

$$\hat{\mathbf{v}}_{vox}(t) = \mathbf{L}_v^T [\mathbf{L}_v \mathbf{L}_v^T]^{(-1)} \mathbf{b}(t)$$

$$\mathbf{G} = \mathbf{L}_v \mathbf{L}_v^T$$

$$\hat{\mathbf{v}}_{vox}(t) = \mathbf{L}_v^T \mathbf{G}^{(-1)} \mathbf{b}(t)$$

G is square (306 x 306) so in principle invertible, but it is rank-deficient, because sensors see the same things

How can we make it invertible?

Regularisation

$$\hat{\mathbf{v}}_{vox}(t) = \mathbf{L}_v^T (\mathbf{G} + \epsilon \mathbf{I})^{(-1)} \mathbf{b}(t)$$

ϵ : a scalar, (higher values indicate more regularisation)

$\mathbf{I}$: the identity matrix

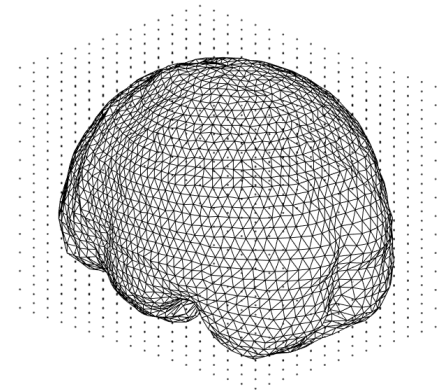
this solution minimises the cost function F :

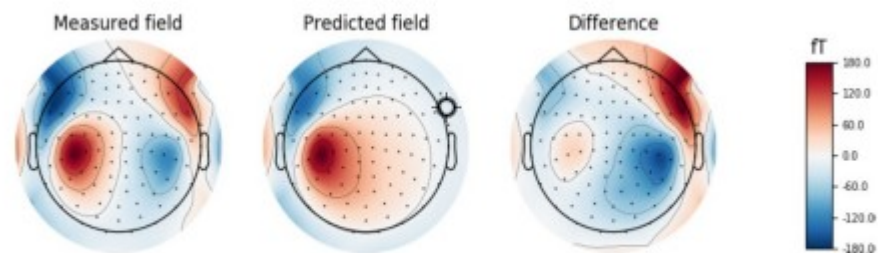
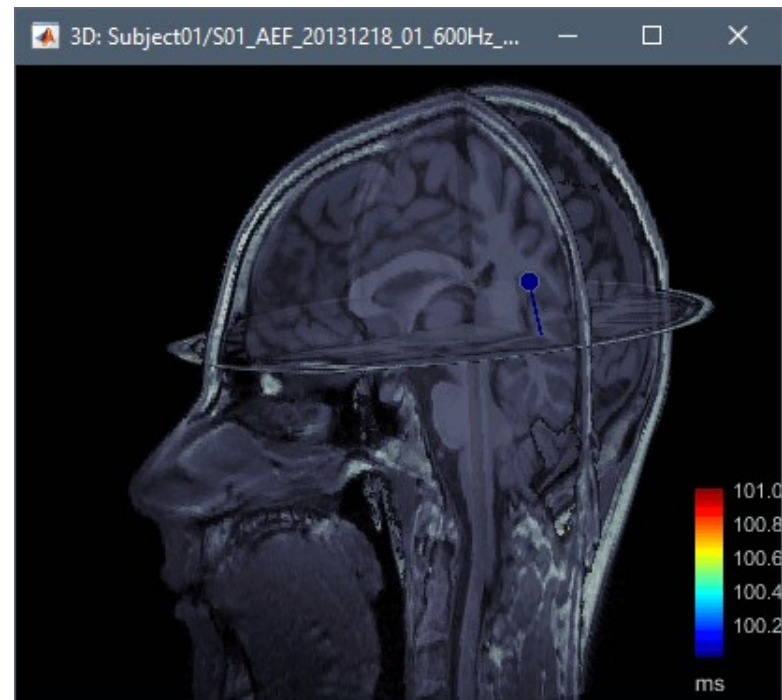
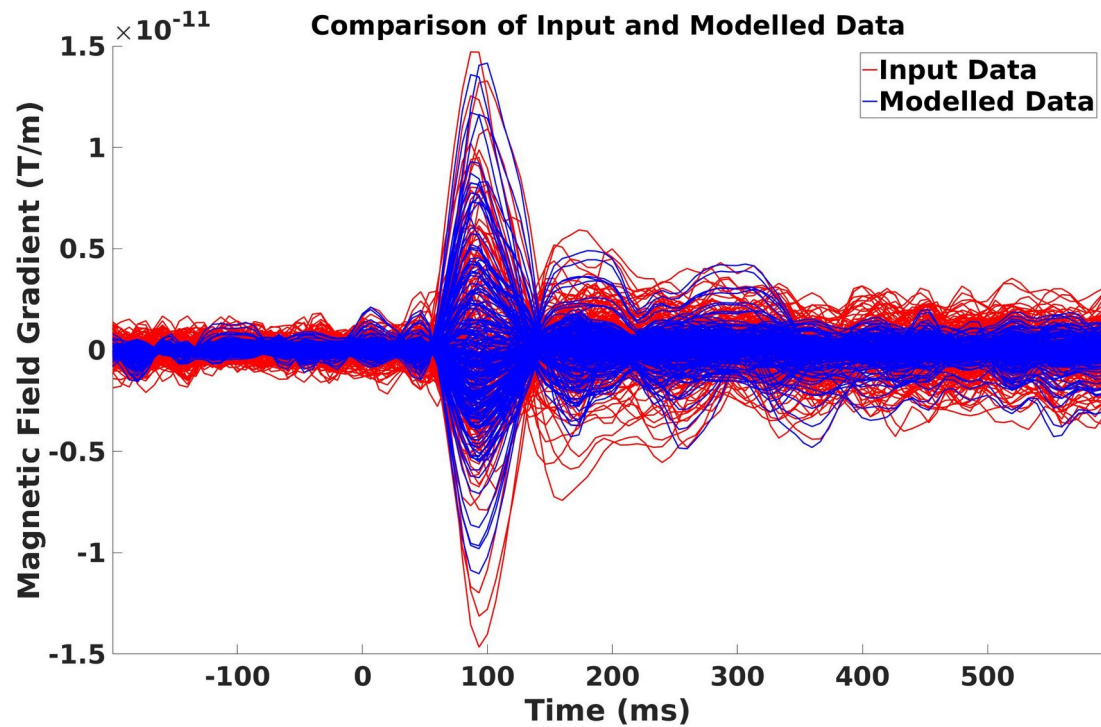
$$F = \|\mathbf{b}(t) - \mathbf{L}_v \hat{\mathbf{v}}_{vox}(t)\|^2 + \epsilon \|\hat{\mathbf{v}}_{vox}(t)\|^2$$

Remaining slides not part of
syllabus; just for reference

Dipole fits

- Two-step procedure for minimizing the error, where the error is the mismatch between MEG data and *inverse* model
 - 1 Scan all the sources within your source space to find the optimal starting position
 - 2 From the optimal starting position, do gradient descent until the error cannot be minimized further.





Dipole fitting

1. Scan all the sources within your source space to find the optimal starting position

Minimize the following expression, i.e. the discrepancy between model and observed data (“+” is the pseudoinverse)

$$[\mathbf{b}(t) - \mathbf{L}(\mathbf{r}) \mathbf{L}(\mathbf{r})^+ \mathbf{b}(t)]^2$$

$\mathbf{L}(\mathbf{r})$ = the lead field for any given source ($\mathbf{r}$)

$\mathbf{b}(t)$ = the observed magnetic field or the electric potential at a given time point (t)

$\mathbf{L}(\mathbf{r})^+$ = the pseudoinverse of $\mathbf{L}(\mathbf{r})$

Derivation

$\mathbf{b}(t)$: measured data

$\hat{\mathbf{s}}(\mathbf{r}, t)$: fitted dipole

$\mathbf{L}(\mathbf{r})$: forward solution

$\hat{\mathbf{b}}(t) = \mathbf{L}(\mathbf{r})\hat{\mathbf{s}}(\mathbf{r}, t)$: according to forward model

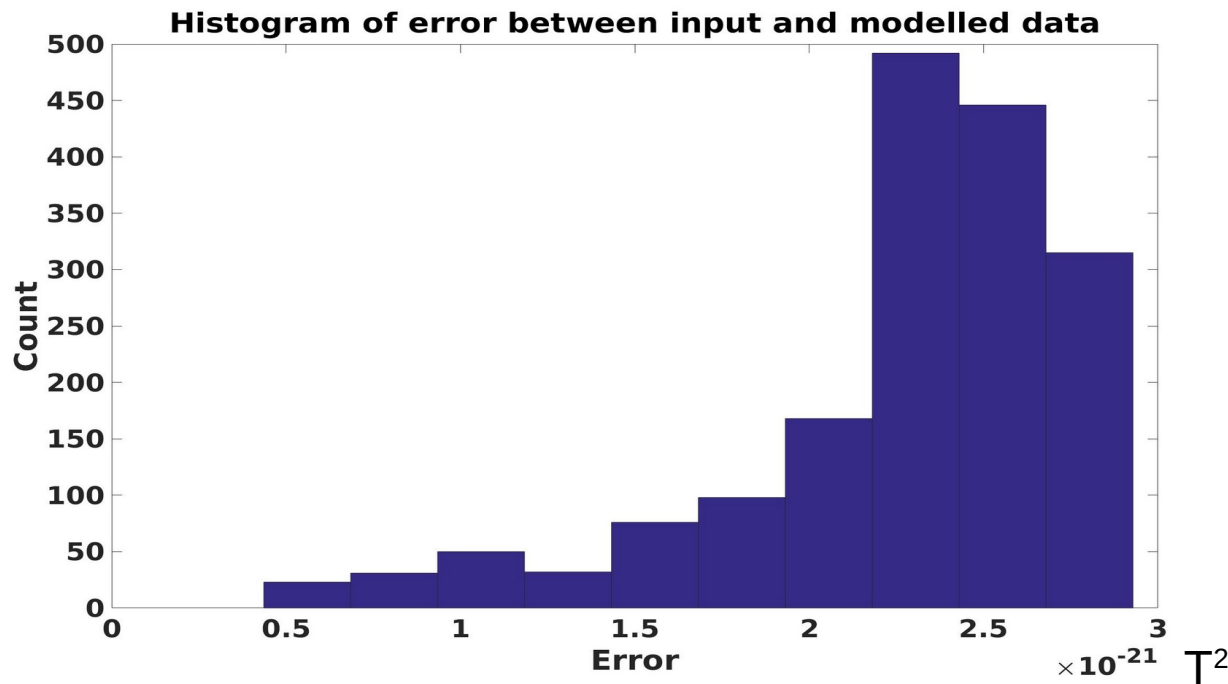
$[\mathbf{b}(t) - \hat{\mathbf{b}}(t)]^2$: to be minimised

$$\hat{\mathbf{s}}(\mathbf{r}, t) = \mathbf{L}(\mathbf{r})^+ \mathbf{b}(t)$$

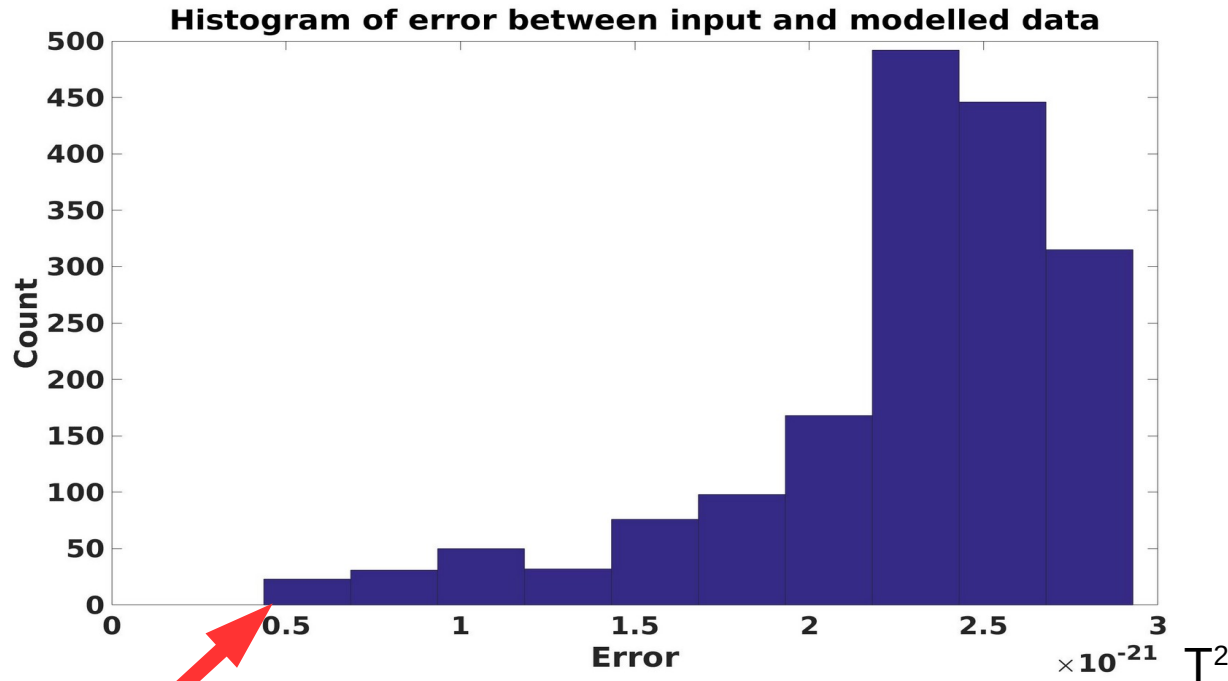
$\hat{\mathbf{b}}(t) = \mathbf{L}(\mathbf{r})\mathbf{L}(\mathbf{r})^+ \mathbf{b}(t)$: using $\hat{\mathbf{s}}(\mathbf{r}, t)$ for $\mathbf{s}(\mathbf{r}, t)$

$[\mathbf{b}(t) - \mathbf{L}(\mathbf{r})\mathbf{L}(\mathbf{r})^+ \mathbf{b}(t)]^2$: cost function

1. Scan all the sources within your source space



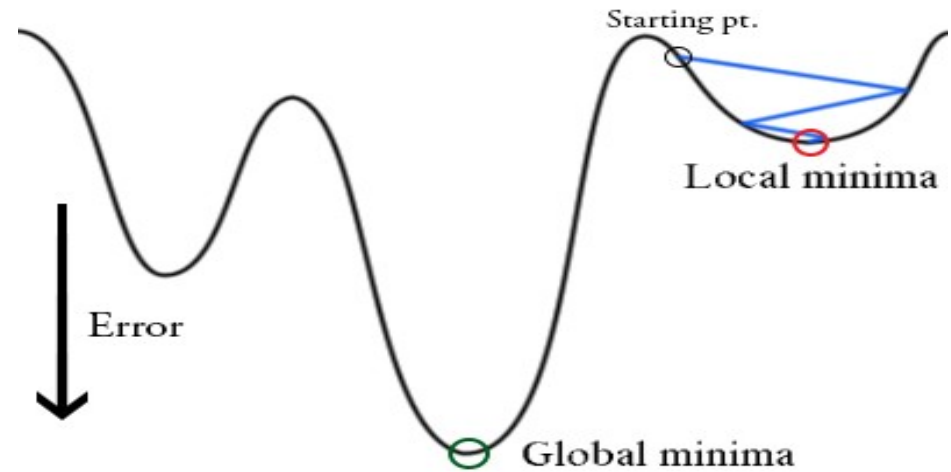
The source position corresponding to the smallest error is used as the starting position



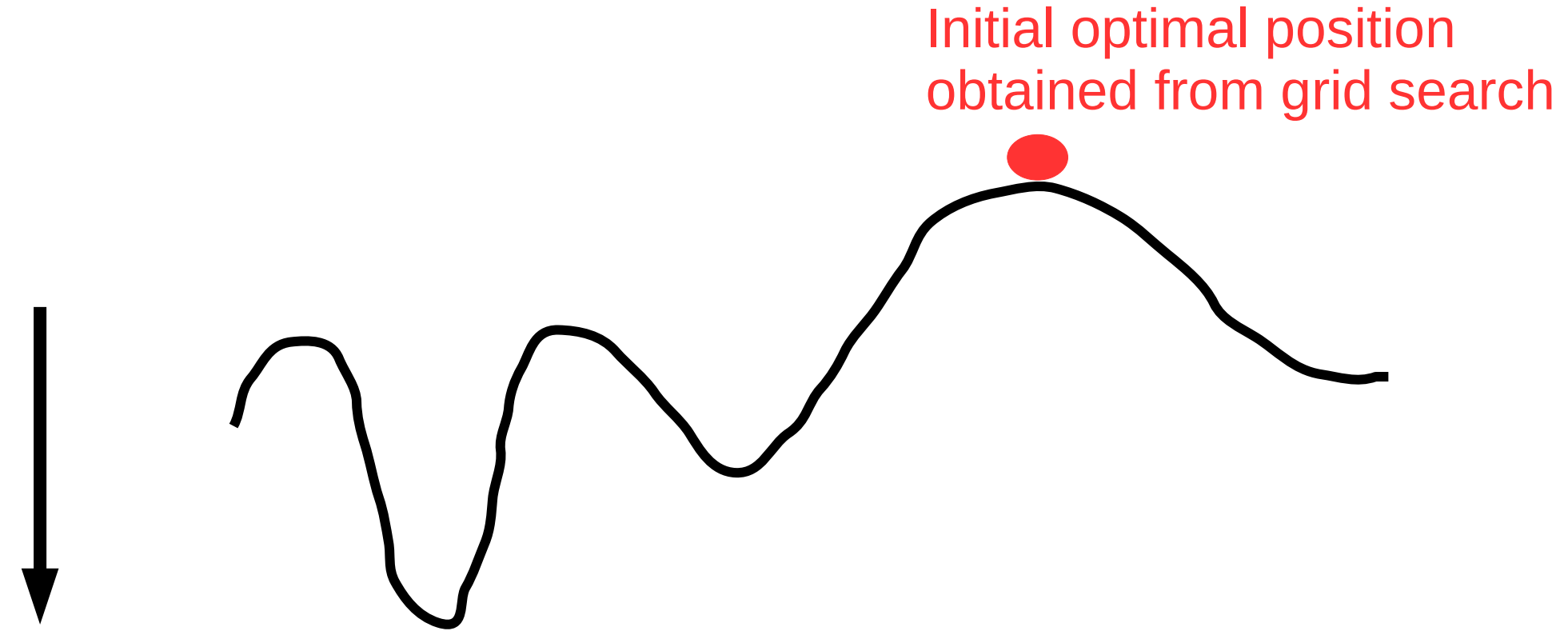
Start from corresponding position

Gradient descent

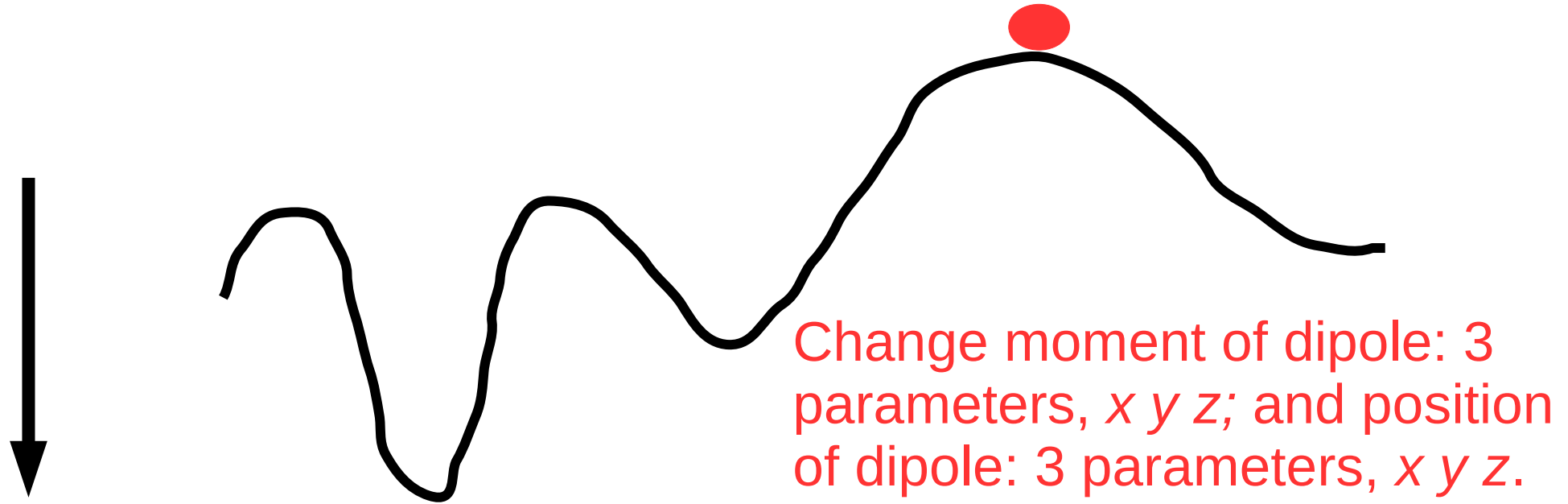
- From the optimal position, do gradient descent until the error cannot be minimized further.
 - The dangers of local minima exist



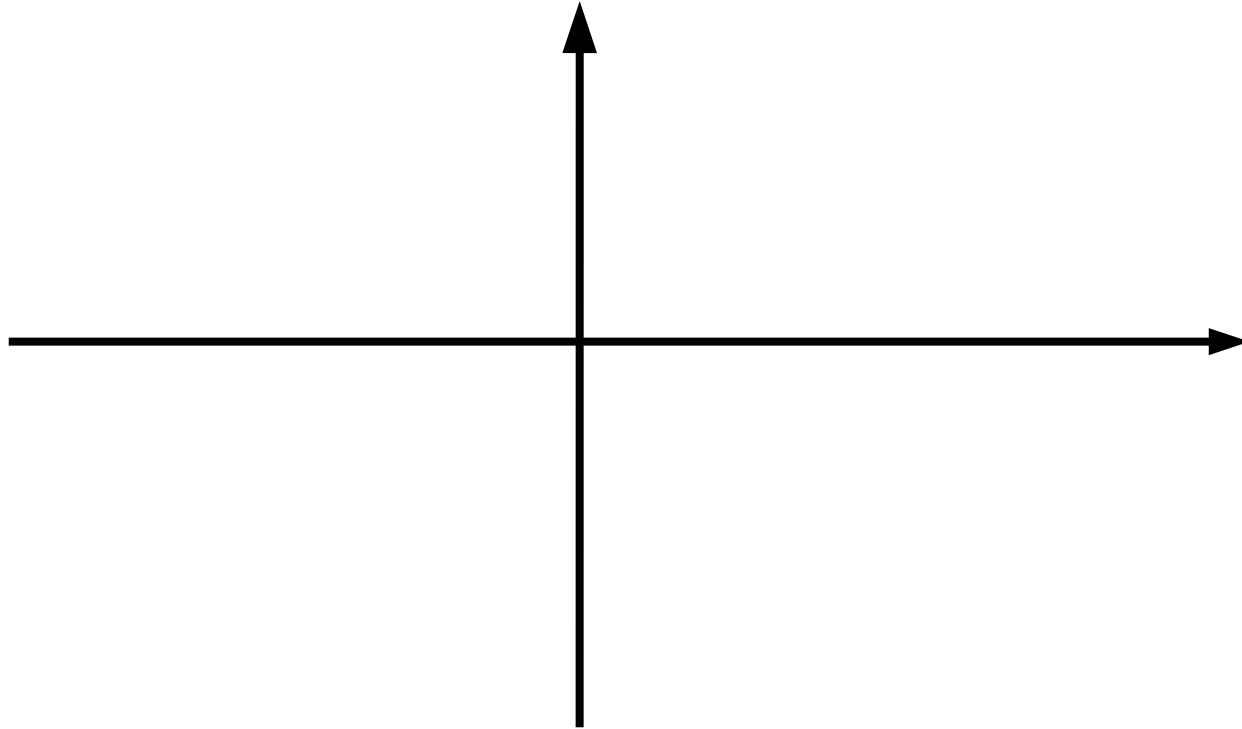
2. Gradient descent



2. Gradient descent

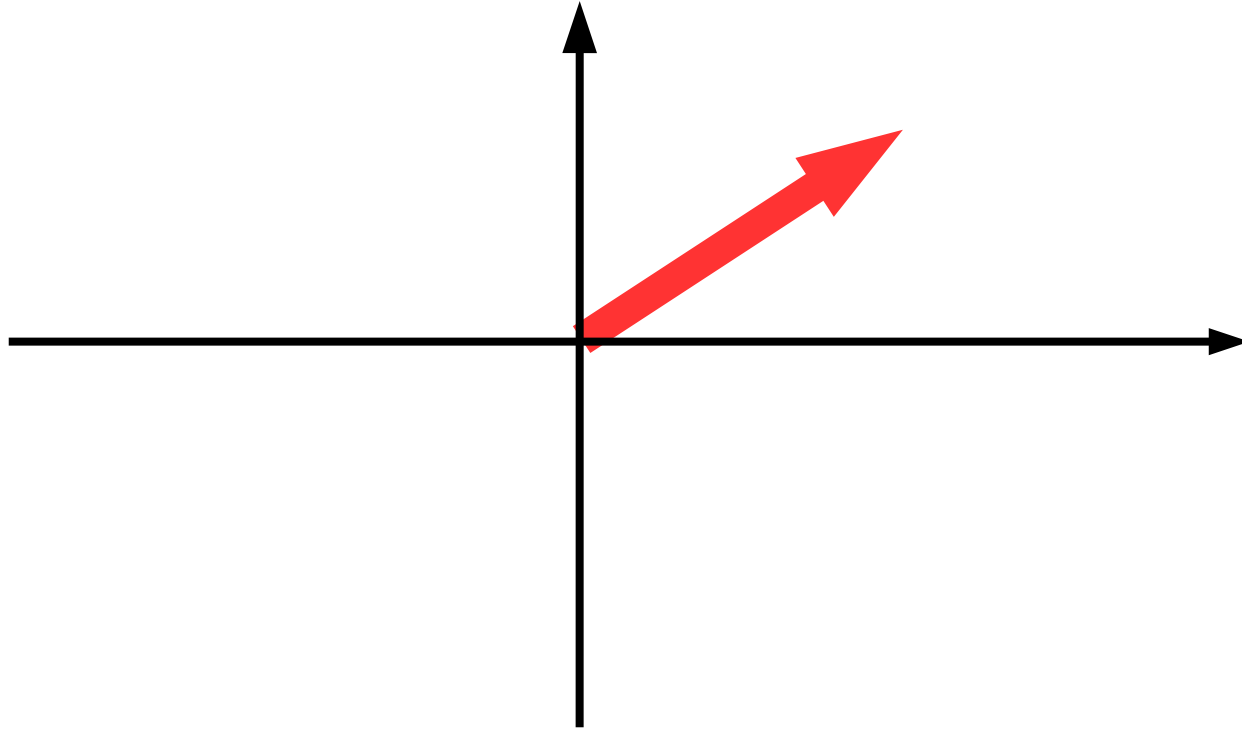


Moment – vector space (example in 2d)

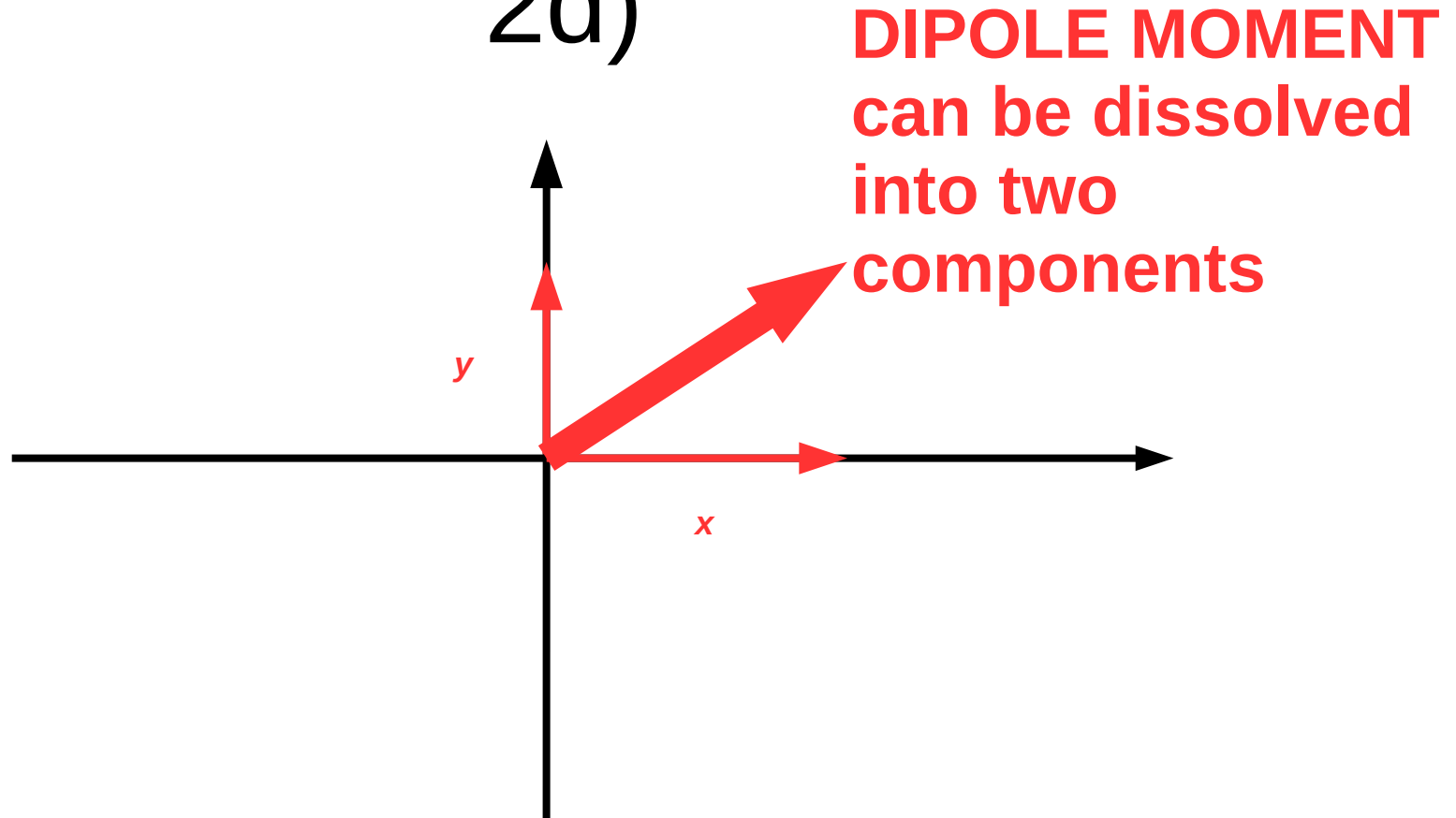


Moment – vector space (example in 2d)

DIPOLE MOMENT



Moment – vector space (example in 2d)

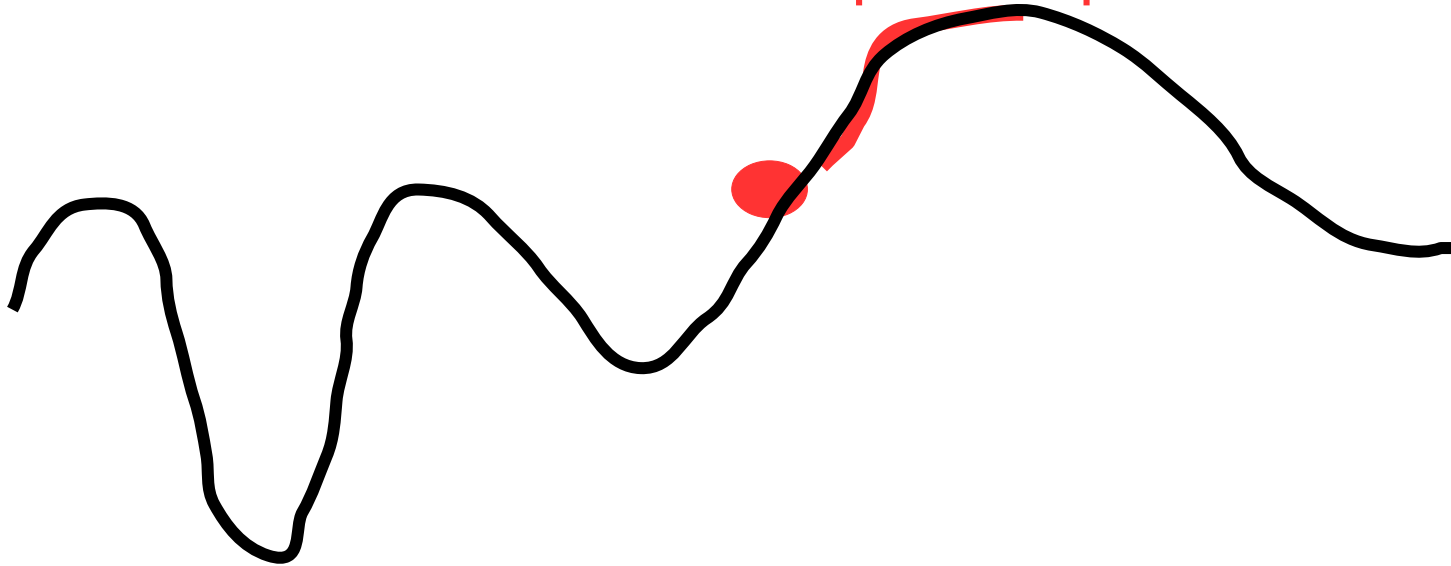


2. Gradient descent

Change moment of dipole: 3 parameters, x y z ; and position of dipole: 3 parameters, x y z .



ERROR

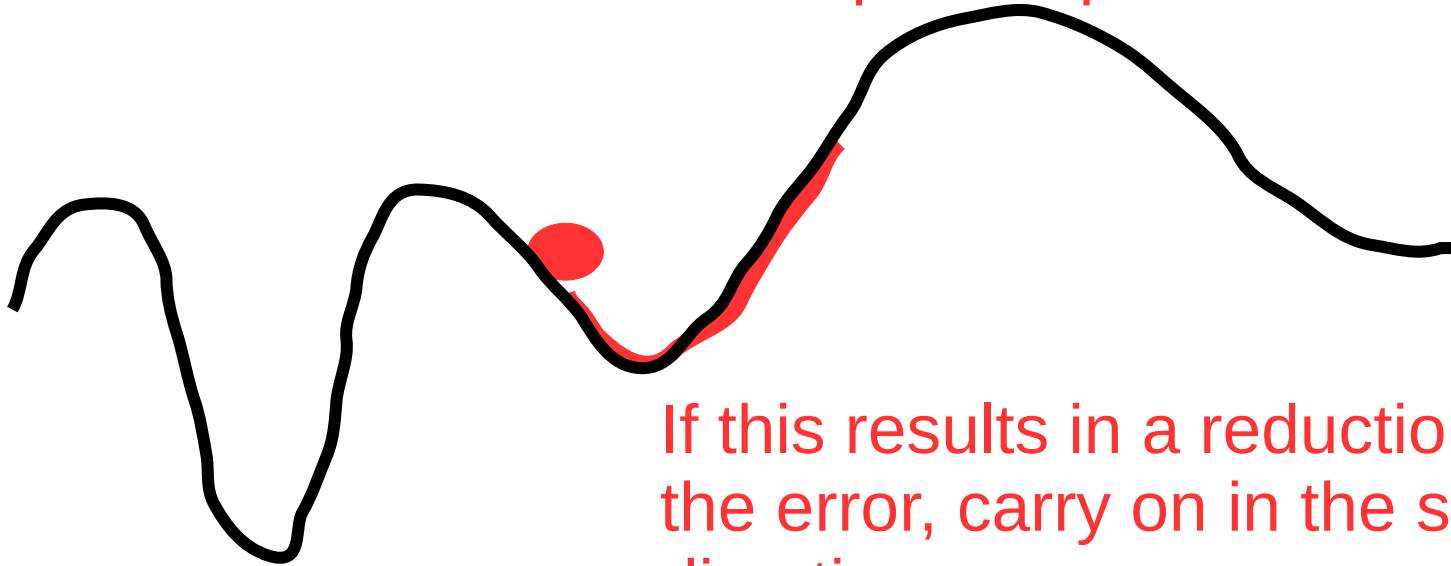


2. Gradient descent

Change moment of dipole: 3 parameters, x y z ; and position of dipole: 3 parameters, x y z .



ERROR



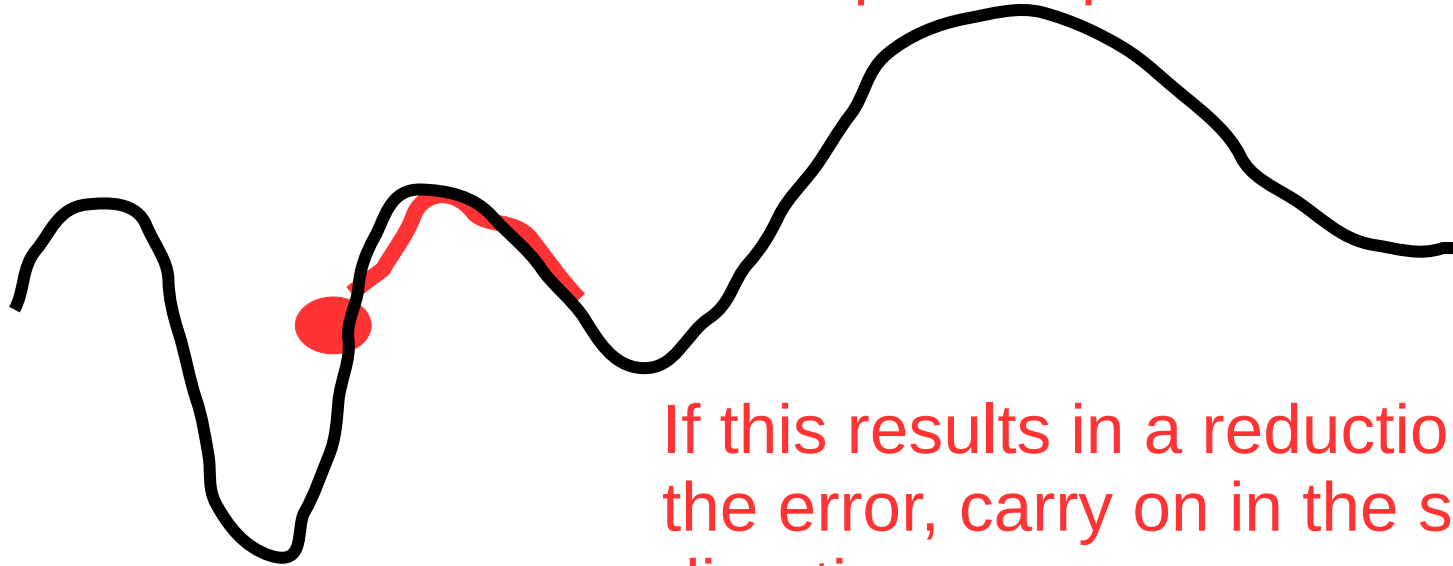
If this results in a reduction of the error, carry on in the same direction

2. Gradient descent

Change moment of dipole: 3 parameters, x y z ; and position of dipole: 3 parameters, x y z .



ERROR



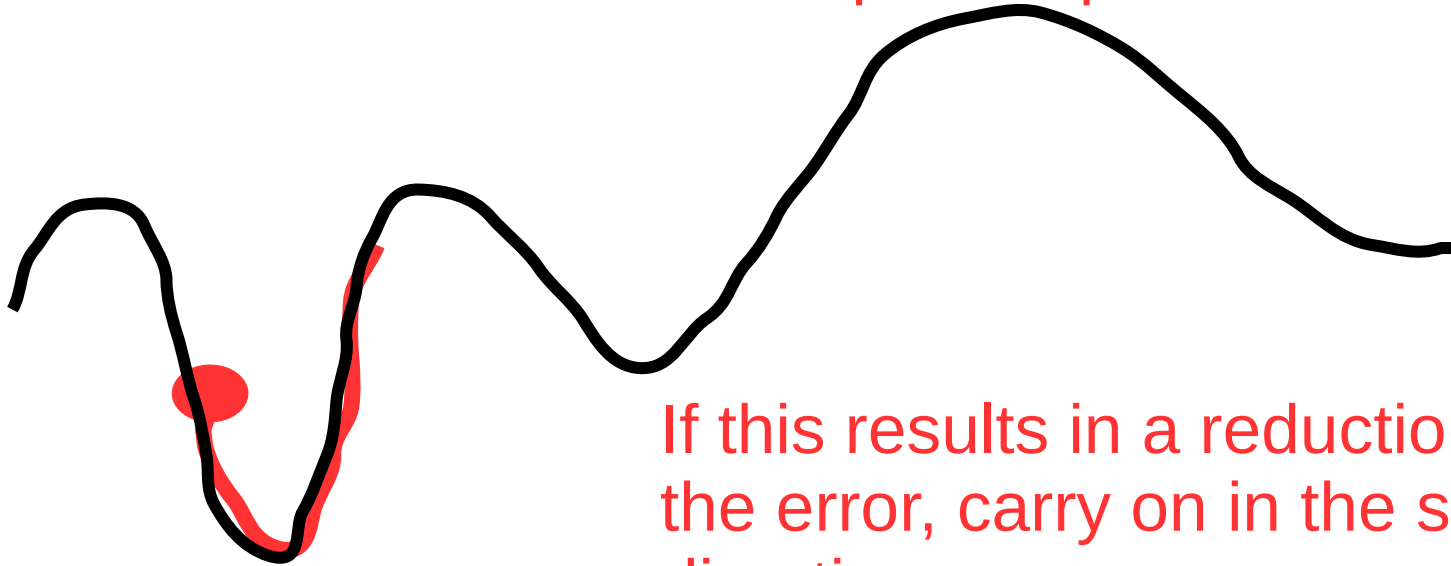
If this results in a reduction of the error, carry on in the same direction

2. Gradient descent

Change moment of dipole: 3 parameters, x y z ; and position of dipole: 3 parameters, x y z .



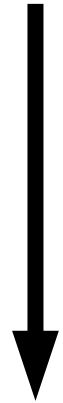
ERROR



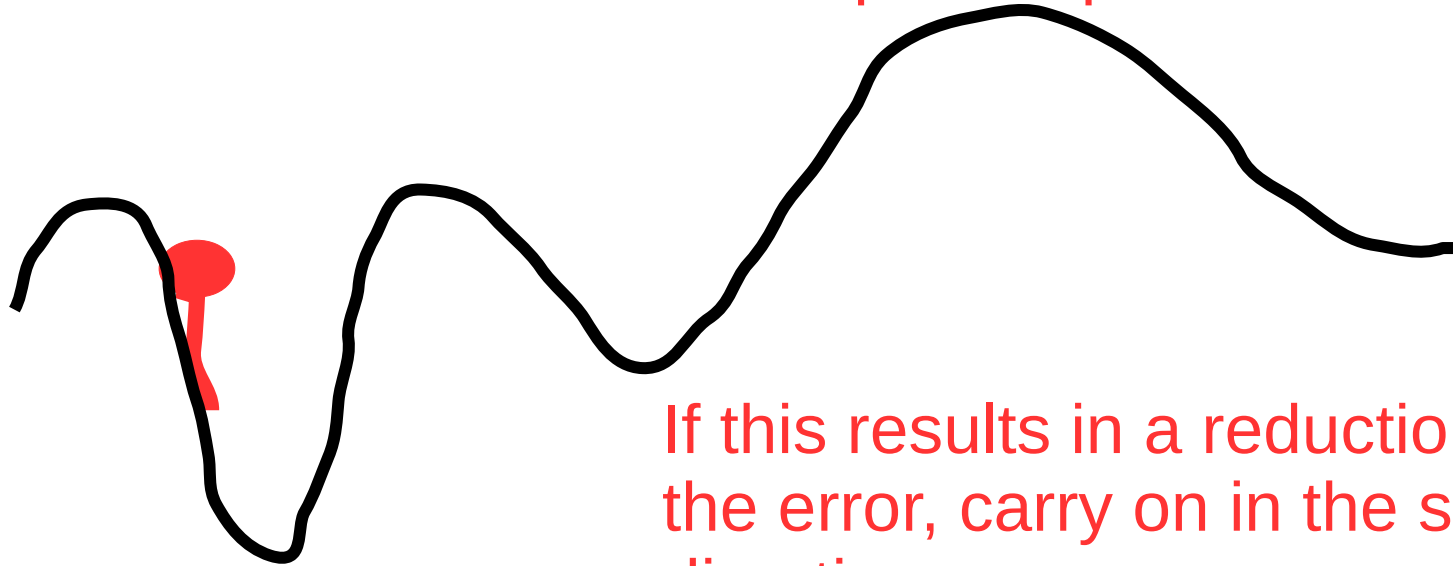
If this results in a reduction of the error, carry on in the same direction

2. Gradient descent

Change moment of dipole: 3 parameters, x y z ; and position of dipole: 3 parameters, x y z .



ERROR



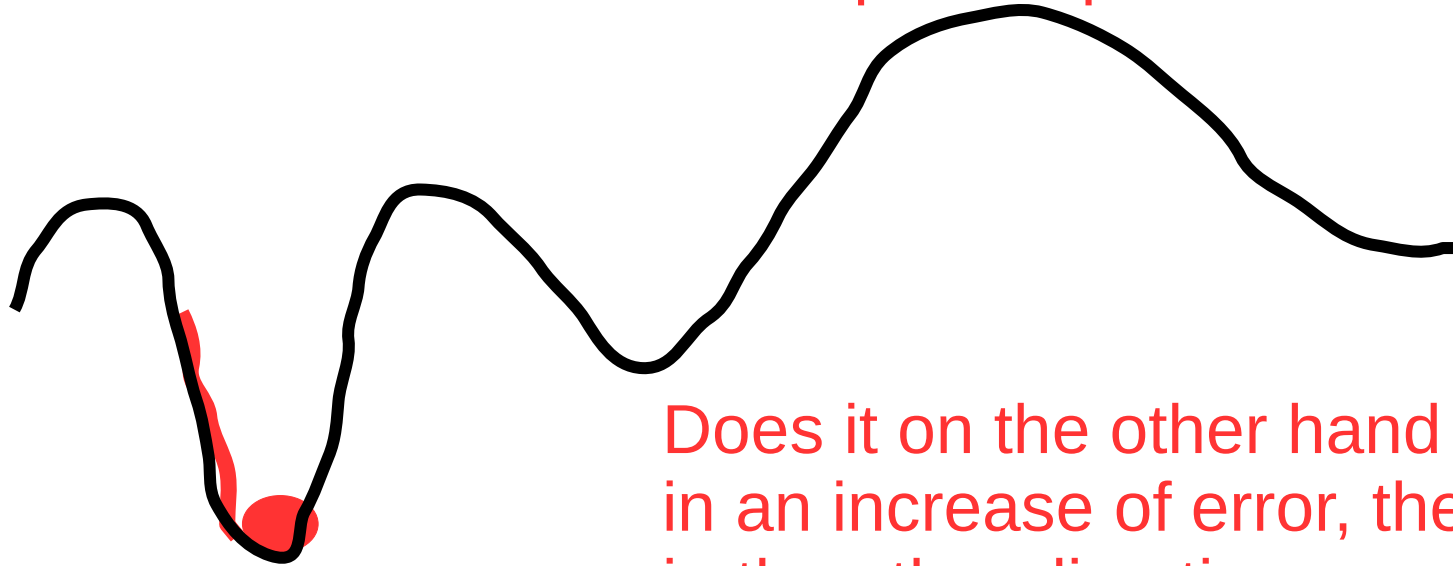
If this results in a reduction of the error, carry on in the same direction

2. Gradient descent

Change moment of dipole: 3 parameters, x y z ; and position of dipole: 3 parameters, x y z .



ERROR



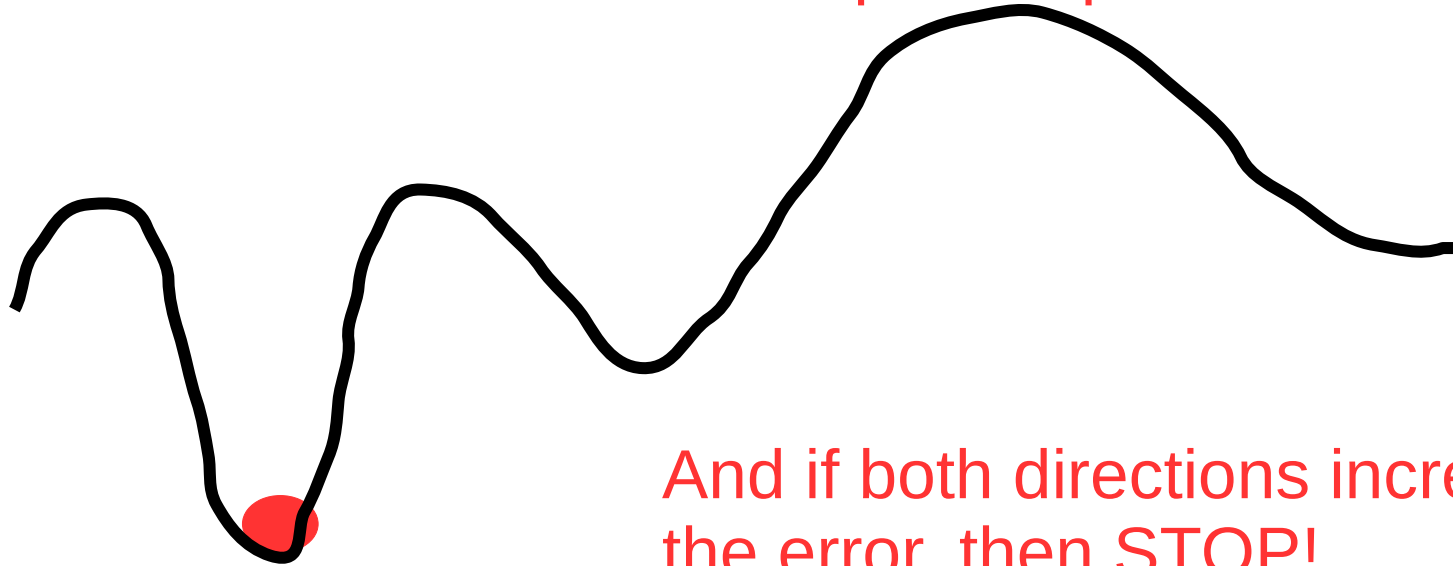
Does it on the other hand result in an increase of error, then go in the other direction

2. Gradient descent

Change moment of dipole: 3 parameters, x y z ; and position of dipole: 3 parameters, x y z .



ERROR



And if both directions increase the error, then STOP!

2. Gradient descent

Change moment of dipole: 3 parameters, x y z ; and position of dipole: 3 parameters, x y z .



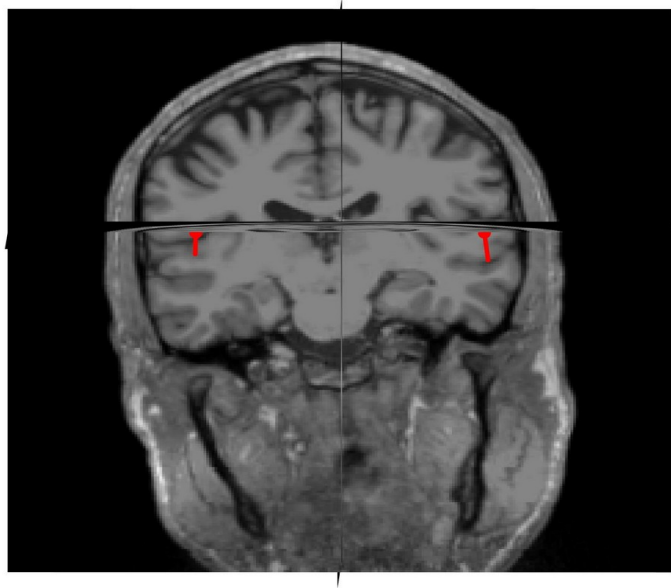
ERROR



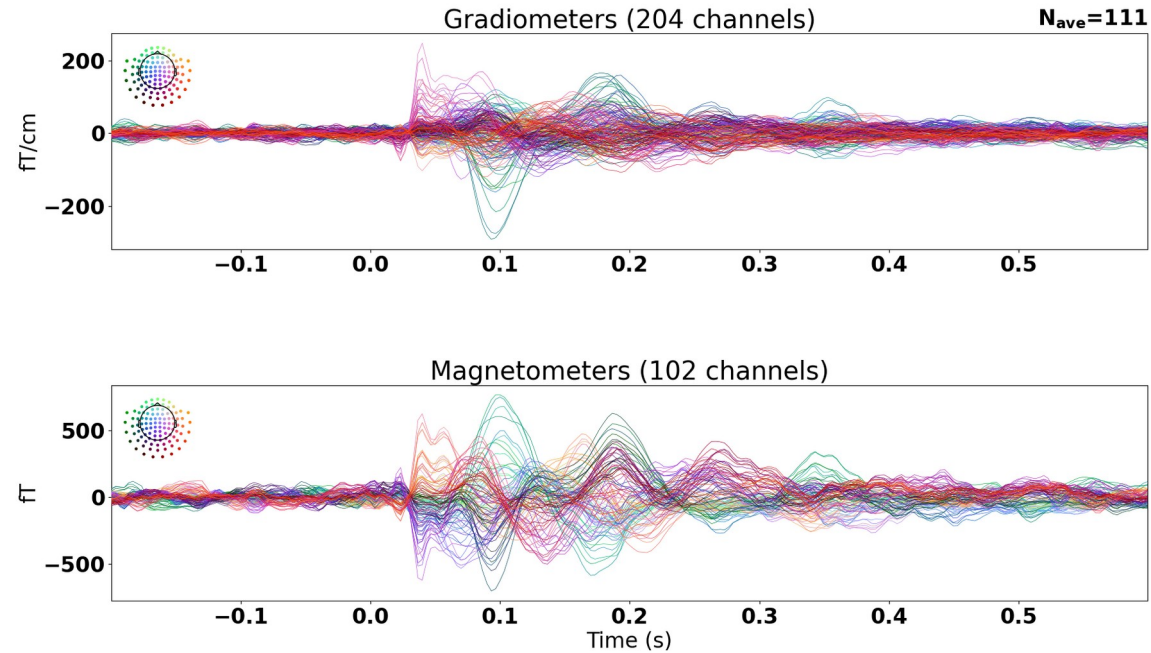
You might end up in local minima though

Dipole fit results

100 ms

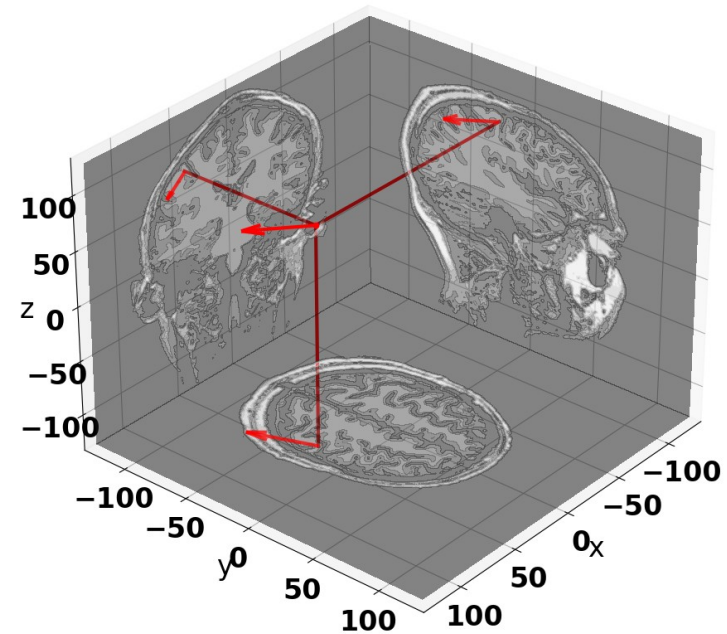
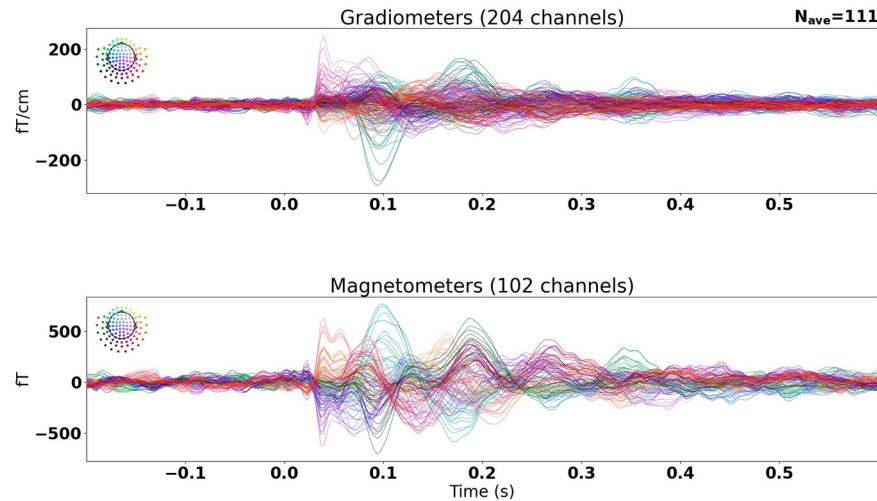


Somatosensory stimulation

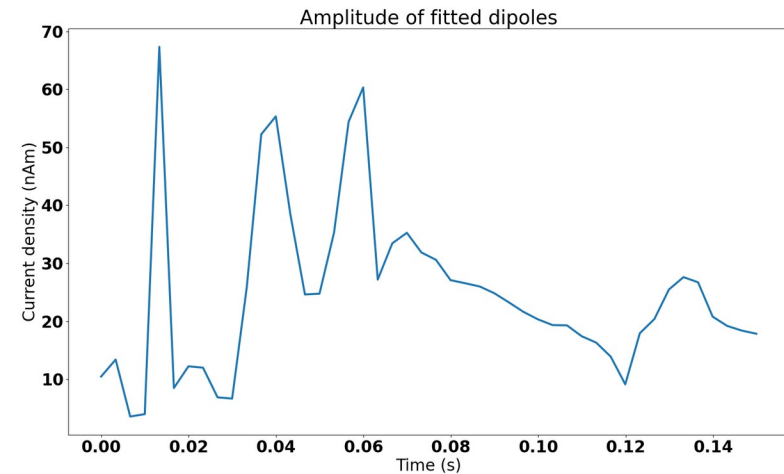
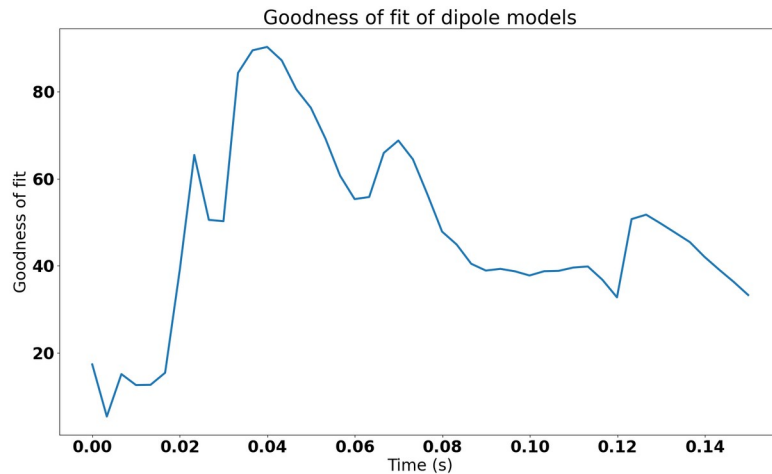


Somatosensory stimulation

Dipole #13 / 46 @ 0.040s, GOF: 90.3%, 55.3nAm
MRI: (43.8, -23.5, 75.6) mm

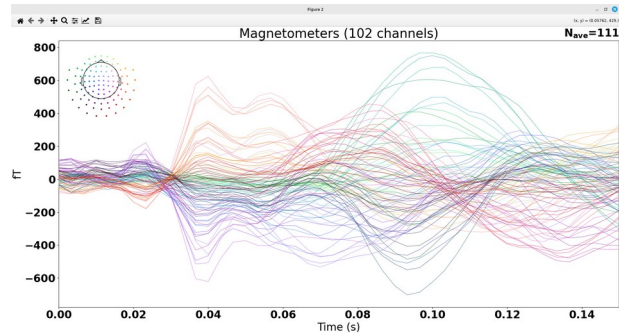


Dipole fit results – goodness of fit

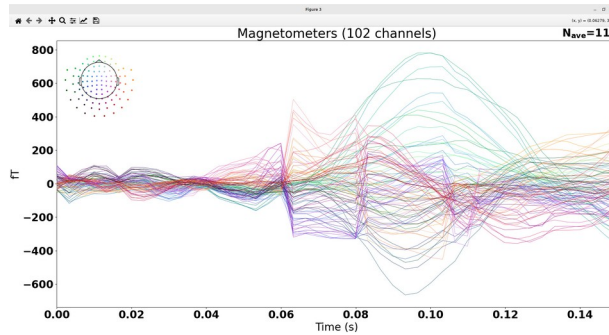
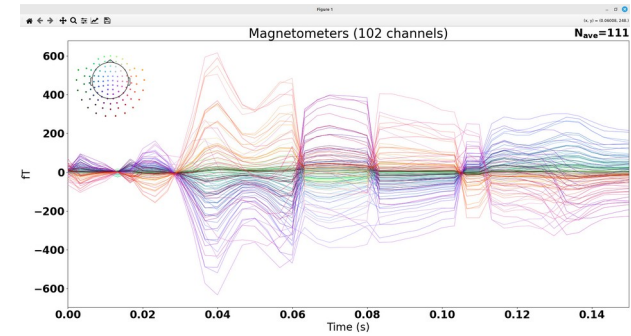


0 = dipolar model explains nothing in the data
100 = dipolar model explains everything in the data

Measured



Model



Residual (measured - model)

CODING EXAMPLE

dipole fitting

may be done in class

Summary

- Dipole fits assume that a single (or few) source(s) can explain (most of) the observed pattern **b** at a given t
- They are especially good for fitting early, sensory components.
- Several dipoles can be fitted by sequentially fitting the residuals

Reading questions

- What does it mean that the minimum-norm is a non-adaptive filter?
- Why is the generalised inverse needed to estimate $\hat{\nu}_{vox}(t)$?
- How is the spatial matched filter similar to dipole fitting?
- What happens with the MNE if data that has a non-brain origin is part of the $\mathbf{b}(t)$?

Next class – fitting dipoles to your own data